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BANK RUNS WITH AND WITHOUT BANK FAILURE

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ABSTRACT

We study the causes and consequences of bank runs. By applying large language models to historical newspapers, we create a comprehensive database of bank runs in U.S. history with information on 3,984 runs on individual banks from 1863 to 1934. Our novel data allow us to establish that runs are considerably more likely in weak banks but also occur in strong banks, especially in response to negative news about the real economy or the broader banking system. However, runs typically only result in failure for banks with poor fundamentals. Strong banks survive runs through various mechanisms, including signaling strength, interbank cooperation, and temporary suspension. At the local level, runs on banks with poor fundamentals translate into substantially larger declines in deposits, lending, and manufacturing activity than runs on strong banks. Our findings imply that poor fundamentals are central to explaining both when runs occur and when they have severe economic effects, tempering the view that small shocks can generate discontinuous jumps to bad equilibria through self-fulfilling run dynamics.

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A Online Appendix is available at http://www.nber.org/data-appendix/w35504

1 Introduction

Bank runs are a salient feature of many financial crises. While it is well understood that financing illiquid assets with demandable liabilities makes banks vulnerable to a sudden withdrawal of funding (Diamond and Dybvig, 1983), the role of runs in financial crises remains controversial. Runs can be seen as the key turning point, whereby even small shocks can generate severe crises with widespread bank failures. Alternatively, runs can be viewed as a consequence of deeper fundamental problems in the financial system, exacerbating crises rather than being their primary cause.

This paper uses novel microdata on bank runs in the U.S. from 1863 to 1934 to study their causes and consequences. We address the following questions: What are the determinants of runs? When do bank runs result in bank failure? What are the real economic consequences of bank runs? Can runs trigger the failure of healthy banks and amplify small shocks into large crises?

A key challenge for understanding the causes and consequences of runs is the absence of systematic microdata on the occurrence of bank runs. Existing micro-level studies of bank distress typically focus on bank failures, as these are objectively recorded in historical records (Calomiris and Mason, 2003; Martin, Puri and Ufier, 2026; Correia, Luck and Verner, 2025). However, this approach misses the broader universe of runs that do not involve failure. At the same time, studies of bank runs typically focus on granular data from specific episodes (Iyer and Puri, 2012; Iyer, Puri and Ryan, 2016; Cipriani, Eisenbach and Kovner, 2024) or consider aggregate data on banking panics (Calomiris and Gorton, 1991; Baron, Verner and Xiong, 2021; Jamilov et al., 2024). Due to the absence of systematic micro data on bank runs, a consensus on how fundamentals map into the likelihood of bank runs and panics has yet to emerge (see Blanchard, 2018).

We overcome this gap by studying the near-universe of runs in the U.S. banking system from 1863 to 1934 as recorded in contemporary newspapers. Our approach leverages advances in large language models (LLMs) and the digital accessibility of historical newspaper articles (Dell et al., 2023). This allows us to construct a novel database of 3,984 runs on individual banks during 1863–1934. We also identify 13,772 bank suspensions and 10,341 failures. Each event we identify is backed by newspaper accounts, which we publicly document at finhist.com/bank-runs. Our approach allows us to study bank runs that result in failure, runs that result in temporary suspension but not failure, and runs that have no further consequences. We combine these data on bank runs with information on bank balance sheets, data on macroeconomic conditions, and newly digitized data on local business failures and manufacturing activity.

We provide three main findings. First, runs are considerably more likely in weak banks, but can also occur in strong banks. Second, runs typically only result in failure for banks with weak fundamentals. Third, runs on weak banks lead to substantially larger local declines in lending and real activity than runs on strong banks. Taken together, our findings suggest that poor fundamentals are key for whether runs pass through into failure and have severe consequences for the broader economy.

The historical U.S. banking system provides an appealing environment to study runs. Our sample covers the numerous panics of the National Banking Era and the Great Depression. Bank runs are less common in contemporary banking systems due to a variety of government interventions that affect depositor behavior, such as deposit insurance (Iyer et al., 2019) and lending of last resort (Metrick and Schmelzing, 2021). However, government interventions in the financial sector were more limited before 1934, making runs much more common in this time period. This historical laboratory has shaped economists' understanding of bank runs and banking crises (e.g., Sprague, 1910; Friedman and Schwartz, 1963; Bernanke, 1983; Calomiris and Gorton, 1991; Wicker, 2006; Richardson, 2007), though this work has largely relied on aggregate data or episode-specific analysis of narrative evidence. Our paper, in contrast, combines both comprehensive narrative accounts and exhaustive bank-level microdata.

We start by validating our novel bank runs database. Spikes in the rate of bank runs recorded in newspapers line up closely with existing narrative chronologies of banking crises in the United States (Jalil, 2015; Baron, Verner and Xiong, 2021), both at the aggregate and regional level. For national banks, the annual unconditional probability of a run is 0.39%. This number rises to 1.4% during crisis years. Moreover, national bank failures for which newspapers recorded a bank run involve a 5-percentage-point larger deposit outflow right before failure than those without mention of a run. For banks that survive bank runs, we also find large deposit declines in their annual financial statement around the event. In addition, our dataset achieves a match rate of about 95% when cross-validating against four ground-truth samples.

With these novel data on runs, we address three fundamental questions on the role of bank runs in bank failures and financial fragility.

First, what are the determinants of runs? We empirically examine the role of bank fundamentals and negative news about the economy in triggering bank runs. We establish that runs are substantially more likely in banks with weak observable fundamentals. We measure fundamentals using balance sheet measures capturing bank capitalization and the ability of a bank to finance itself with cheap deposits relative to expensive noncore funding. Existing research shows that these metrics capture bank health and failure risk

(White, 1984; Calomiris and Mason, 2003; Correia, Luck and Verner, 2025). We document that runs are considerably more likely in poorly-capitalized banks, banks with higher leverage and less liquid assets, and banks that rely more on noncore funding.

While runs are more likely in banks with weak fundamentals, the ability to predict runs with fundamentals alone is somewhat low—especially compared to the predictability of bank failures. However, the incidence of runs is well explained when combining information on bank fundamentals with information on the local and national economic and banking conditions. For instance, we find that runs are more likely after a downturn in the stock market, a rise in local non-financial business failures, or runs on other banks in the same city. News shocks that are not bank-specific help explain runs on healthier banks.

Second, when do bank runs result in bank failure? We find that a bank's annual probability of failure increases by 38 percentage points when subject to a run. This is a large effect, given that the annual unconditional probability of bank failure is 0.89%. However, it also implies that a run is not a death sentence. There are more runs without bank failure than runs with failure, highlighting a key novel insight from our approach.

We further establish that whether a run results in failure depends crucially on a bank's financial health. Runs are significantly more likely to result in failure for banks with weak fundamentals. Conditional on a run, a bank with very weak fundamentals—defined as being in the lowest decile of our preferred metric that summarizes a bank's financial health—has a 59% probability of failing. In contrast, the *ex ante* strongest banks, measured as being in the top decile of the same metric, seldom fail when subject to a run. Thus, weak bank fundamentals are necessary for a run to be associated with bank failure.

To better understand how banks with sound fundamentals survive runs, we leverage additional context provided in newspaper articles. We find that newspapers often report that surviving banks accommodate withdrawals and attempt to signal strength to depositors. A common signal of strength is an equity or deposit injection by owners or other investors. In some cases, banks weathered liquidity shocks by borrowing from other banks. Further, 28% of runs that do not result in failure nevertheless lead to a temporary suspension of convertibility. Suspension was a common way to successfully resolve runs, often alongside bank examination to signal solvency to depositors. In 11% of runs, newspapers mention that the local bank clearinghouse is involved by providing liquidity or conducting an examination.

The finding that runs can occur for both weak and strong banks, but that only weak banks fail from runs, has important implications for theories of bank runs. First, the finding that runs happen predominantly in weak banks supports the notion that fundamentals are important (Morris and Shin, 2000; Goldstein and Pauzner, 2005). Second, the fact that even healthy banks can be subject to runs following adverse news shocks is consistent with information-based theories of bank runs, where depositor confusion leads to withdrawals from both weak and strong banks because depositors struggle to distinguish which banks are solvent (Gorton, 1988; Chari and Jagannathan, 1988; Calomiris and Gorton, 1991; Dang, Gorton and Holmström, 2020). Finally, the fact that both weak and strong banks can be subject to a run even though they can be separated based on simple public balance sheet metrics suggests that depositors are at times inattentive. The predictability of both runs and failures suggests that depositors tend to be sleepy and are slow in responding to weak bank fundamentals (Hanson et al., 2015). However, the importance of news shocks suggests that salient information can cause depositors to wake up and run.

In line with this interpretation, we further find that runs following negative public signals have contrasting implications for bank failure, depending on the nature of the signal. Runs following runs on other local banks are associated with a lower probability of failure. This suggests news of other runs makes depositors "jittery," leading to a rise in runs on healthy banks that do not result in failure. In contrast, runs following a rise in the local business failure rate are associated with a higher probability of failure, consistent with these runs responding to a rise in bank losses that erodes solvency.

We also explicitly examine the scope for non-fundamental runs to trigger failure. We define non-fundamental runs as those that, according to the newspapers, are caused by misinformation or depositor confusion and are not related to bank or economic conditions. These runs are the most plausible empirical equivalent to the pure liquidity-driven runs studied theoretically in Diamond and Dybvig (1983). With the assistance of LLMs, we identify a sample of 53 non-fundamental runs on national banks. This exercise also provides sharper identification by zooming in on a set of runs that are relatively random and thus free of confounding factors. The probability of failure following a non-fundamental run is 11%, substantially lower than for the average run. Moreover, we find that these non-fundamental runs trigger failure when they happen to occur in weak or fraudulent banks, reinforcing the key role of poor fundamentals for the pass-through of runs to failure. While strong fundamentals reduce the probability of failure after a run generally, this is especially true for non-fundamental runs for which only banks with the weakest fundamentals fail. This pattern casts doubt on a strong form of the view that liquidity problems alone can trigger severe financial distress.

Third, what are the economic consequences of bank runs with and without failure? At the bank level, we find that banks that are subject to runs and survive see a contraction in deposits and loans of about 7% of pre-run assets. The contraction is persistent, lasting for over four years. To refine the identification, we study the consequences of runs we classify as non-fundamental. Such runs also have modest negative effects on deposits and lending. Further, bank-level deposit and lending contractions following runs are much stronger for banks with *ex ante* weak fundamentals, while the effects are small and not significant for strong banks. Thus, bank fundamentals shape the severity of the consequences of runs, even when runs do not translate into failure.

Finally, we study the consequences of bank runs for local city-level deposits, lending, and manufacturing activity. We first show that runs only lead to declines in deposits, lending, and manufacturing activity when they occur in *ex ante* weak banks. Runs on weak banks translate into a 5% decline in a proxy of city-level manufacturing production. Non-fundamental runs entail no significant adverse local effects. Further, we use the richness of our bank distress episodes data to consider the city-level consequences of runs *without* failure versus runs *with* failure. We find that spikes in the local rate of runs *without* failure lead to modest declines in deposits and loans at the city level and no effect on manufacturing activity. In contrast, runs with failure lead to more severe contractions in local deposits, credit, and manufacturing activity.

Taken together, our findings have two broad implications. First, while runs can occur in both weak and strong banks, poor fundamentals are necessary for runs to result in bank failures. Second, failures, and their corresponding weak fundamentals, are a key mechanism for banking sector distress to translate into substantial contractions in credit and real activity. Pure liquidity events, in which depositors run due to fear or confusion but bank fundamentals are sound, are not associated with severe economic outcomes at the local level. These patterns, in turn, temper the view that small shocks can result in large jumps to bad equilibria via self-fulfilling runs on demandable debt. Rather, bank runs are only associated with major economic distress when they take place in the context—and as a consequence—of already weak fundamentals. At the same time, because even insolvent banks can remain liquid for some time, a run may be an important determinant for the timing and economic costs of bank distress.

Related Literature Our paper relates to a rich empirical literature on bank runs, bank failures, and financial crises (see, e.g., Sufi and Taylor (2021) and Frydman and Xu (2023) for overviews).

The main contribution of our paper is to provide insights from a novel comprehensive database that covers nearly four thousand bank runs in the historical U.S. banking system. Our novel data allows us to provide new systematic evidence on which banks become subject to bank runs, which shocks trigger runs, which banks fail in runs, how banks survive runs, and the consequences of runs with and without failure for broader credit and real activity. Hence, our study relates to micro-level studies of bank runs. Existing evidence suggests that deposit growth is lower in relatively weaker banks (see, e.g., Chen et al., 2024) and especially when solvency concerns are salient during crisis episodes (Schumacher, 2000). Jiang et al. (2023), Metrick (2024), and Cipriani, Eisenbach and Kovner (2024) provide evidence that the banks most exposed to interest rate risk were those most affected by the banking stress during the March 2023 banking turmoil. Furthermore, previous work has shown that banks can withstand runs when they have strong ties to their depositors (Iyer and Puri, 2012). Runs by relatively less well-informed depositors are also less likely to result in failure (see, e.g., OGrada and White, 2003; Saunders and Wilson, 1996; Calomiris and Mason, 1997; Iyer, Puri and Ryan, 2016; Blickle, Brunnermeier and Luck, 2024; Jamilov et al., 2026). Moreover, existing work suggests that interbank and wholesale markets may be crucial for how runs play out (Afonso, Kovner and Schoar, 2011; Perignon, Thesmar and Vuillemey, 2018; Mitchener and Richardson, 2019). Finally, Richardson and Troost (2009) also show that central bank interventions can be important in influencing bank survival rates during times of heightened financial distress.

Our paper also relates to studies of banking panics using aggregate or regional data. Gorton (1988), Calomiris and Gorton (1991), Alston, Grove and Wheelock (1994), Wicker (1996), and Wicker (2006) present either time-series or regional evidence suggesting that U.S. banking panics from the National Banking Era through the Great Depression were largely predictable responses to fundamental shocks that altered depositors' perceptions of bank solvency. Baron, Verner and Xiong (2021) argue that widespread panic runs are not a necessary feature of banking crises, showing that panics are typically preceded by declines in bank equity prices that reflect the recognition of underlying losses. Jamilov et al. (2024) identify systemic bank runs with narrative evidence and observed deposit outflows and find that such runs occur infrequently, but when they do, they are associated with large, persistent output losses.

Our paper also relates to a large literature on understanding bank failures (e.g., White, 1984; Calomiris and Mason, 2003; Correia, Luck and Verner, 2025). These studies find that failed banks displayed weak balance sheets before failure and argue that runs usually accelerated the failure of already insolvent banks, rather than causing widespread collapse of otherwise solvent institutions; see Correia, Luck and Verner (2026) and references therein for a comprehensive review. Our paper makes a distinct contribution to this literature by expanding the scope of bank distress to systematically include runs and suspensions without bank failures. We provide the novel insight that runs can occur in strong banks, but they typically only result in failure for weak banks.

Further, our paper contributes to the literature that shows that financial disruptions can have persistent consequences for the real economy by linking the consequences of both bank failures and bank runs without failures to real economic outcomes at the bank and local level. Bernanke (1983) emphasizes the role of bank failures and the resulting credit market impairments in deepening the Great Depression. Chodorow-Reich (2014) provides micro-level evidence that credit supply shocks during the 2008–09 financial crisis led to substantial employment losses among firms dependent on weak lenders. Similarly, Frydman, Hilt and Zhou (2015) document that runs on trust companies during the Panic of 1907 curtailed investment and dividend payments by connected firms, illustrating how financial distress transmits to the real economy even outside traditional banks.

Finally, our methodology provides an innovation to the narrative approach to identification in macroeconomics and finance (e.g., Romer and Romer, 2004). By providing historical classifications of banking crises, the narrative approach has been fruitful for understanding crises (Bordo et al., 2001; Reinhart and Rogoff, 2009; Jordà, Schularick and Taylor, 2013; Baron, Verner and Xiong, 2021). The narrative approach has generally been used in a macroeconomic context with a relatively limited set of events. Our methodology using textual analysis and LLMs allows us to apply the narrative approach at a scale that would previously have been insurmountable without a squadron of researchers. Furthermore, the narrative approach to classifying banking crises involves researcher judgment, leading to some disagreement about which episodes constitute widespread bank runs (e.g., Jalil, 2015; Baron, Verner and Xiong, 2021; Hoon et al., 2025). Our newspaper-based approach provides a systematic method to resolve these disagreements based on large-scale and consistent analysis of a given corpus of text. It is particularly well-suited to understanding events such as runs when such events are recorded in the text, as it usually does not require judgment to determine whether there is a discussion of a bank run.

Roadmap The next section outlines testable predictions from theories of bank runs. Section 3 describes the methodology behind the construction of our novel bank runs dataset. Section 4 validates our database and documents the incidence of bank distress in the U.S. from 1863 to 1934. Section 5 studies the empirical determinants of bank runs. Section 6 studies the pass-through of runs to failure. Section 7 estimates the consequences of runs for affected banks and for local deposits, lending, and manufacturing activity. Section 8 concludes.

2 Conceptual Framework

To guide our empirical analysis, we outline different prominent theories of the causes of bank runs and their predictions for whether runs result in bank failure. We then discuss the predicted real economic consequences of bank runs with and without failure.

Our emphasis on bank failure as a central outcome is motivated by a long tradition in macro-finance arguing that bank failure is particularly costly for real activity because it disrupts credit intermediation (e.g., Bernanke, 1983). Bank lending is "special" because it mitigates informational and agency frictions through relationship-based screening and monitoring. When a bank fails, this informational capital is destroyed, leading to persistent reductions in credit. As a result, bank failure provides a natural mechanism through which runs translate into broader real economic costs. Nevertheless, as we discuss below, runs that do not result in failure may still impose economic costs through alternative channels.

2.1 Causes of Runs and Implications for Bank Failure

To fix ideas, we use a simple framework to outline the empirical predictions from three different theories of runs: non-fundamental bank runs (e.g., Diamond and Dybvig, 1983), fundamental-based panic runs (e.g., Morris and Shin, 2000; Goldstein and Pauzner, 2005), and information-based runs (e.g., Gorton, 1988; Chari and Jagannathan, 1988). We summarize the main insights from this framework in the main text, leaving the full model details and derivations to Appendix B.

Setup There are two dates, $t \in \{1, 2\}$. There is a bank that initially holds liquid cash C and illiquid loans L, financed by deposits D and equity E. There is a continuum of risk-neutral depositors, each holding one unit of deposits. Deposits are demandable at $t = 1$. However, the bank lacks the liquid funds to pay all depositors in the initial period, $D > C$. In $t = 2$, the loan portfolio pays off θL. We refer to the gross return θ as the fundamental of the economy. There is no discounting, deposits pay zero interest, and the return on cash is zero.

At time $t = 1$, depositors observe θ and decide whether to keep their deposits in the bank or withdraw. Depositors who keep their deposits in the bank until $t = 2$ receive a convenience benefit of c per unit of deposit. If total withdrawals exceed cash, the bank is illiquid, and it is forced into failure (receivership). In failure, the loan portfolio value is reduced to $(1 - \rho)\theta L$. The reduction in value ρ can reflect that the receiver is less skilled at managing bank assets than the banker or a fire sale discount. We assume sequential service of withdrawals. Depositors who withdraw while the bank is liquid receive full repayment, whereas depositors remaining in the bank in failure receive a pro-rata share of the remaining assets.

Non-fundamental Runs The bank is fundamentally insolvent when the fundamental is sufficiently low that it cannot fully repay depositors, $\theta < \theta^S \equiv \frac{D-C}{L}$. However, the bank may be subject to a run that renders it illiquid and forces it into failure if $\theta < \theta^L \equiv \frac{D-C}{(1-\rho)L}$. The threshold for run-induced failure exceeds the solvency threshold, $\theta^L > \theta^S$. Thus, for $\theta \in [\theta^S, \theta^L)$, a self-fulfilling run constitutes an equilibrium. Because depositors are served on a first-come, first-served basis, they run if they expect others to run, leading to a self-fulfilling panic. A run that renders a bank illiquid will also render it insolvent by forcing it into value-destroying failure.

Thus, the bank is exposed to the possibility of a self-fulfilling panic run that is not strongly tied to bank or broader economic fundamentals. We refer to these as *non-fundamental runs* (Diamond and Dybvig, 1983).[1] Non-fundamental runs can occur in relatively healthy banks and following a small shock that brings the fundamental θ just below θ^L. These runs are thus unpredictable jumps to a bad equilibrium that are not necessarily linked to particularly weak fundamentals.

Fundamental-based Panic Runs Models of non-fundamental runs do not predict a sharp relation between fundamentals and runs. However, runs may be more likely following bad realizations of fundamentals. Models of *fundamental-based panic runs* capture this idea (Morris and Shin, 2000; Goldstein and Pauzner, 2005).[2] In Goldstein and Pauzner (2005), bank fundamentals are stochastic, but agents do not have common knowledge about fundamentals. Instead, each depositor receives a slightly noisy private signal of the bank's asset value θ. This global games set-up can feature a unique equilibrium as a function of fundamentals. A bank run occurs when bank fundamentals θ fall below a specific threshold $\theta^* \in (\theta^S, \theta^L)$. Importantly, the model generates panic runs, in the sense that, absent the run, the bank would have been solvent and survived for $\theta \in [\theta^S, \theta^*)$.[3]

[1] Many papers explore non-fundamental runs in a Diamond-Dybvig framework (e.g., Peck and Shell, 2003). He and Manela (2016) model the dynamics of information-acquisition in response to rumor-based runs, showing that socially inefficient information acquisition can increase the likelihood of runs. Their model also captures the notion of gradual withdrawals.

[2] See also Allen and Gale (1998) and Rochet and Vives (2004) for models where runs occur after the realization of poor fundamentals. To capture the idea that runs should occur in relatively weaker banks, Ennis and Keister (2003) and Gertler and Kiyotaki (2015) assume that non-fundamental runs are more likely in banks with lower θ.

[3] In practice, depositors may be sleepy and not attentive to bank fundamentals (e.g., Egan et al., 2025). Therefore, depositors may react slowly to weak fundamentals, leading to runs on weak banks once these

Runs thus amplify bad fundamentals. The empirical prediction is that fundamental-based panic runs should occur in banks with weak fundamentals.[4]

Information-based Runs In another class of theories, runs are modeled as a response to aggregate public signals that lead depositors to revise their assessment of the riskiness of deposits. We refer to these as *information-based runs*. These aggregate signals are not bank-specific. This class of models assumes that bank assets are opaque and depositors cannot tell apart good and bad banks (Dang et al., 2017). In particular, suppose at time $t = 1$ depositors receive a noisy public signal $y = \theta + \epsilon$, where ϵ is a mean-zero shock. In this setting, if a signal is sufficiently bad, $y < \overline{y}$, a run constitutes an equilibrium. Such runs away from bank debt can occur even in response to small shocks (Gorton, 2012; Gorton and Ordoñez, 2014).[5] Information-based theories also capture the idea that depositors can misinterpret public signals and mistakenly run on healthy banks. If the realization of the shock ϵ is sufficiently low, then a bank with a high $\theta > \theta^L$ can be run.

Depositors can react to a range of signals. For example, depositors might run in response to a signal that a recession is imminent, such as following a decline in the stock market, a rise in business failures, or runs on other banks (Gorton, 1988; Calomiris and Gorton, 1991). Depositors might misinterpret a long line at a bank as a signal that the bank is insolvent, even though the line was caused by a liquidity demand shock (Chari and Jagannathan, 1988). Related to this, runs can be driven by herding (Gu, 2011).[6] The bank run in the film *Mary Poppins* is an evocative example. Runs on other banks may lead depositors to revise their beliefs that their own bank is insolvent and run, an information-based contagion (Chen, 1999). Behavioral theories of flawed belief formation can also lead to runs on healthy banks, as suggested by the original connotation of "panic." Information-based theories predict that panics should occur in response to adverse public signals, leading to runs on both weak and strong banks.

Prediction 1. ***Causes of bank runs:*** *Non-fundamental runs can happen both in banks with weak and strong fundamentals. Fundamental-based panics happen in weak banks. Information-based*

banks are already insolvent (Correia, Luck and Verner, 2025).

[4]In Calomiris and Kahn (1991) and Diamond and Rajan (2001), runs are triggered by negative information about bank assets. In these frameworks, demandable deposits are the optimal contract to discipline bankers.

[5]In recent work by Dang, Gorton and Holmström (2020), runs arise when an adverse signal about the collateral backing bank debt leads agents to produce information about bank debt. As a result, there is a regime switch, and bank debt moves from being information-insensitive to information-sensitive. Agents flee bank debt out of fear that others have private information about the value of bank debt.

[6]Gu (2011) writes a model of herding by depositors, where there can be both efficient and inefficient runs. Inefficient runs occur when depositors are misled by past observed withdrawals through an informational cascade.

runs are more likely in weak banks, but can also happen in strong banks after negative news that is not necessarily bank-specific.

Bank Runs and Bank Failure What do theories of bank runs predict about bank failure? In standard theories of non-fundamental and fundamental-based panic runs, in equilibrium, runs necessarily trigger failure. For example, in Diamond and Dybvig (1983) and Goldstein and Pauzner (2005), the run equilibrium leads to the liquidation of the bank. If the liquidation can be avoided, then there is no run in the first place. Thus, these models predict that, in equilibrium, runs cannot occur without failure.[7]

In contrast, information-based theories allow for runs without failure, as depositors may run on strong banks following overly negative signals. Several mechanisms can prevent the failure of a solvent bank in response to a run, even in the absence of a public lender of last resort. Runs might be managed by accommodating withdrawals or attempting to signal that the bank is solvent, such as through information provision or equity injection. For instance, the run in the celebrated and commonly invoked (see, e.g., Royal Swedish Academy of Science, 2022) film *It's a Wonderful Life* is resolved without failure because George Bailey is able to restore depositor confidence through a mix of persuasion and use of his personal wealth.[8] A run on a healthy bank by uninformed depositors can also be managed by interbank cooperation. In the National Banking Era, bank clearinghouses would regularly inspect member banks and provide liquidity in response to runs (Timberlake, 1984; Gorton, 1985b). Heavy runs can be resolved by suspension of convertibility, whereby banks refuse to convert deposits into currency for a time. Gorton (1985a) models suspension of convertibility as a costly way for a solvent bank to signal its solvency and avoid an inefficient liquidation. If these mechanisms are empirically relevant, then they should imply that even if runs on solvent banks do occur, they should rarely lead to bank failures.[9]

In our simple framework, suppose that if withdrawals exceed cash, the bank can

[7]This statement is based on a version of global games models in which all depositors receive almost the same signal about the state of the world, and which is the most commonly considered version. Note that partial runs that do not result in failure are possible in models such as Goldstein and Pauzner (2005) when the common knowledge assumption is more broadly relaxed and when depositors do not receive the exact same signal, i.e., the noise does not go to zero.

[8]Note that George Bailey operates a Building & Loan rather than a commercial bank. The run in the movie is fended off after George Bailey uses both his ability as a gifted speaker and his honeymoon savings to end the run.

[9]Note that information-based runs where agents make mistakes can be resolved with suspension, but they still occur in equilibrium. However, in full-information models, such as classic models of non-fundamental runs, suspension of convertibility would rule out inefficient runs on solvent banks in the first place.

choose one of two actions. Either it can enter failure, where assets are liquidated at $(1-\rho)\theta L$, as above. Or, it can suspend and pay a proportional cost τ to have its books examined and certify that it is solvent. Supposing that the cost of examination is lower than that of failure, $\tau < \rho$, the bank chooses to temporarily suspend and pay the cost τ if $\theta > \theta^{Susp} \equiv \frac{D-C}{(1-\tau)L}$, where $\theta^{Susp} \in (\theta^{S}, \theta^{L})$. Thus, if depositors see an overly negative public signal of fundamentals, but true fundamentals are relatively strong, $\theta \geq \theta^{Susp}$, then the bank is subject to a run, suspends, is examined, and reopens, thereby avoiding failure. A mechanism such as suspension and examination thus reduces the set of solvent banks that can be pushed into failure by a run. Moreover, because survival is more likely, the run threshold itself falls, making runs less likely, especially on strong banks.

Prediction 2. ***Bank runs and failures:*** *In canonical non-fundamental and fundamental-based run models, run equilibria typically imply failure. In information-based models, runs can occur on solvent banks and need not end in failure when mechanisms such as suspension and examination, or interbank support are available.*

2.2 Broader Economic Consequences of Runs and Failures

Our analysis speaks to two broad conceptual views about the consequences of runs for the banking system and the real economy. One view is that the run itself is a key discontinuous turning point whereby a small shock can translate into high economic costs through deposit outflows, bank failures, and credit contraction. This view plays a key role in prominent accounts of the Great Depression and Great Recession (Friedman and Schwartz, 1963; Gorton, 2012; Bernanke, 1983, 2018). Theories of non-fundamental runs and fundamental-based panic runs most directly embody this mechanism (Diamond and Dybvig, 1983; Goldstein and Pauzner, 2005). Information-based theories also allow runs to lead to discontinuous jumps, but the consequences of runs depend more on underlying fundamentals and on whether banks can survive the run.

Another view is that runs themselves are not the root cause of severe downturns, but are instead primarily a reflection of deeper fundamental solvency problems (Admati and Hellwig, 2014; Gennaioli and Shleifer, 2018; Baron, Verner and Xiong, 2021). Solvency problems are the cause of severe economic contraction, as in models of the bank lending channel (Holmstrom and Tirole, 1997; Gertler and Kiyotaki, 2010). Bank capital is the key state variable capturing the ability of the banking system to intermediate funds. If runs often arise in response to these fundamental weaknesses, they may be the trigger that results in inevitable bank failures and economic costs, but not their root cause. In this view, avoiding runs would not prevent the costs of poor fundamentals. At the same time, even if runs are not the root cause of banking sector distress, runs may be an important amplification mechanism for bad fundamentals, leading to worse outcomes than if the run were averted (e.g., Goldstein and Pauzner, 2005; Gertler and Kiyotaki, 2015).

While these two views are challenging to fully distinguish, they entail different empirical implications. On the one hand, if runs themselves are key turning points and the primary cause of distress, then runs should unconditionally be associated with severe outcomes, irrespective of a bank's initial fundamentals.[10] On the other hand, if runs themselves are not the root cause of distress, runs should only be associated with severe consequences when they result in failure. In that case, runs should result in bank failure and large economic contractions when they occur in banks with weak fundamentals, but they should be less costly for banks with sound fundamentals.

Finally, we note that, even absent outright bank failure, runs may have adverse real effects by reducing the quantity and increasing the cost of bank funding, thereby increasing the cost of credit intermediation. Theoretical models of the *bank lending channel* capture the idea that financing constraints among borrowers and banks imply real effects of shocks to bank funding or capital (Holmstrom and Tirole, 1997).[11] Again, in cases where banks survive a run, bank-lending channel models predict that the cost of these runs is higher in weaker, less well-capitalized banks. Our analysis will thus compare runs with and without failure in weak and strong banks to test this idea.

Prediction 3. ***Real consequences of runs and failures:***

(a) If runs themselves are key turning points and the primary cause of distress, then runs should typically be associated with severe outcomes, irrespective of initial bank fundamentals.

(b) If runs are symptoms or amplifiers of deeper fundamental banking sector distress, they should result in severe economic consequences when they occur in banks with weak fundamentals, but be less costly if they occur in banks with sound fundamentals.

3 Data

3.1 Novel Bank Distress Events Database

We construct a novel and comprehensive dataset of bank run events recorded in newspapers for all U.S. banks. The dataset is available at finhist.com/bank-runs and is searchable

[10]Of course, observing severe outcomes after runs does not imply that the run was the *cause* of these outcomes, so this is a necessary but not sufficient condition.

[11]A large empirical literature finds evidence for this channel (e.g., Peek and Rosengren, 2000; Khwaja and Mian, 2008; Chodorow-Reich, 2014; Frydman, Hilt and Zhou, 2015; Huber, 2018).

by year, state, and city. The full bank distress events database comprises events between 1800 and 1963. For this paper, we focus on the 1863–1934 sample to span the period from the start of the National Banking Era to the introduction of federal deposit insurance.

Concepts and Definitions Our conceptual definition of a bank run is an episode in which a substantial number of depositors suddenly demand that their bank convert claims into cash or the liabilities of another bank. We also collect newspaper mentions of other key bank events: suspensions, reopenings, and failures. Our definition of a suspension is when a bank "closes its doors." This includes both temporary and permanent closures of a bank.[12] A reopening occurs when a bank temporarily suspends and subsequently resumes business. A bank failure is a suspension that becomes permanent.[13] This includes receiverships for national banks and some state banks, and assignments for private and other banks. We exclude orderly voluntary liquidations from failures. For our analysis focusing on national banks, we define failures as national bank receiverships.

Sources We rely on several databases of historical newspapers. Our main source is *Chronicling America* (Library of Congress, 2025), a large-scale database of publicly available historical newspaper scans maintained by the U.S. National Digital Newspaper Program. To conduct our analysis at the article level, we draw on the *American Stories* dataset from Dell et al. (2023) to separate pages into articles.[14] Because copyright limitations restrict the number of articles available in the *Chronicling America* database after the mid-1920s,[15] for the 1923-1935 period we further supplement it with articles from commercial sources.[16] Figure C.2 shows the number of articles used in our final analysis across time and source.

[12] Our definition of suspensions does not include partial suspensions of convertibility, such as when a bank imposes maximum withdrawal limits for each depositor or legally invokes 30/60 day notice rules for savings or time deposits. We found that defining suspension as a bank closing its doors was the most objective and empirically verifiable definition. However, below we separately study whether banks partially suspended payments in response to runs.

[13] In rare cases, banks are placed in receivership but restored to solvency later. We nonetheless classify these cases as failures.

[14] *Chronicling America* provides scanned newspaper images together with metadata for each newspaper issue, as well as text obtained by applying reduced-quality optical character recognition (OCR) techniques. However, its key limitation for our purposes is that it provides content only at the page level and cannot distinguish between different articles. Moreover, the low quality of the text makes it challenging to apply textual analysis or even keyword matching methods. To address this, we use the *American Stories* dataset (Dell et al., 2023). This dataset provides, for most of the scanned pages in *Chronicling America*, layout information indicating the coordinates of each article, as well as text with an OCR of sufficiently high quality for keyword matching.

[15] The number of articles in *American Stories* drops from 7.3 million articles in 1922 to 2.6 million articles in 1923 and only 1.3 million articles by 1930 (see Figure C.1).

[16] In particular, we use Newspapers.com and NewspaperArchive.com, two major commercial newspaper databases, as well as the New York Times' TimesMachine digital archive.

Methodology We produce our dataset of bank distress events in two stages. The procedure is summarized in Figure C.3. In the first stage, we process scanned newspaper images and obtain an intermediate dataset of articles related to bank-distress events. This stage has four components. In the first component, illustrated in Figure C.4, we perform a layout analysis of each page, segment pages into articles, and obtain each article's text through inexpensive but relatively imprecise OCR techniques. Next, we search each article with keywords and regular expressions (see Table C.1), and create a tentative list of articles related to bank distress events. Third, we filter out obvious false positives with an off-the-shelf inexpensive LLM.[17] At this point, we have reduced the total number of newspaper articles by 99.8 percent (from 372 million to 801 thousand), so we are able to apply a more precise analysis. In the fourth and last step, we apply state-of-the-art OCR to obtain high-quality article text, and employ a more sophisticated LLM to (i) determine if the article is really about a bank run or other form of distress, and (ii) extract structured information from each article, including every date, event, and bank discussed. This results in an intermediate dataset of 269 thousand articles discussing 314 thousand bank-distress events (see Figure C.6 for some examples).

At this stage, the dataset has the drawback of being at the article-bank-event level rather than at the bank-event level. Thus, a given bank run might be discussed multiple times throughout many articles. For instance, the failure of the Knickerbocker Trust Company in 1907 is discussed in over 1,000 articles. Further, a single article might not be enough to understand the nature of a bank distress episode, as it might assume some degree of knowledge from earlier articles.

To address this, in the second stage, we transform the intermediate article-bank-event dataset into our final dataset at the bank-episode level. This is done in roughly two steps. First, we validate and standardize state, city, and bank names. This allows us to group all articles for a given bank and to combine this data with datasets containing bank balance sheet information, receivership records, and city- or state-level economic indicators. Appendix C.2 discusses how we standardize state and city names. This step is relatively straightforward and correctly matches more than 99 percent of non-missing records against existing city-level datasets. However, standardizing bank names is considerably more challenging, as there is no comprehensive list of bank names across bank types (national, state, private) or historical periods (Antebellum, National Banking Era, Federal Reserve). To overcome this challenge, we build a new near-exhaustive bank list based on existing bank-level datasets and several newly digitized sources.[18] We then

[17]See Listing C.1 for the LLM prompt used and Figure C.5 for some examples of articles filtered out.

[18]This new list of banks augments those of Weber (2006), Jaremski and Fishback (2018), and Correia,

write a Python package that flexibly matches bank names while being robust to typos and variations on how these names are written.[19] This allows us to match 75% of all bank names between 1863 and 1934, and up to 94% if we focus only on national banks.[20]

In the final step, for each bank, we group all the events that take place within a period of time of up to 365 days. These collections of *events* capture what we call "bank distress *episodes*," which we analyze by feeding the corresponding newspaper articles to an ensemble of more advanced LLMs.[21] An example of a bank distress episode involving three bank events is a bank run, suspension, and reopening. We ask the LLMs to jointly analyze all newspaper articles that describe a given bank distress episode, reclassify all the details of this episode, including event types and dates, and finally classify the type of the overall episode depending on its internal dynamics.[22] To minimize classification errors, we combine queries to independent LLMs, and apply a final manual review.

Episode Types Our bank distress episodes can be interpreted as a self-consistent list of events that affected a specific bank. As mentioned above, we classify bank run, suspension, failure, and reopening events. We then further classify bank distress episodes into the following categories based on their internal dynamics.

First, we can thus study episodes with runs but no failure, distinguishing between those resolved with and without suspension:

- *Run only.* A bank run takes place, but the bank remains open throughout the run, and the run resolves on its own. The bank does not suspend or fail.
- *Run-suspension-reopening.* A bank run takes place. As a consequence of the bank run, the bank suspends and stops paying depositors. However, the bank eventually reopens and does not fail.

Next, we can consider runs with failure:

Luck and Verner (2025), with additional banks manually compiled for this paper, based on *Rand McNally Bankers Directory* (multiple years), *Trust Companies of the United States* (multiple years), and other sources.

[19] Appendix C.3 discusses this package in more detail.

[20] We do not truly know if a bank is a national bank unless we can match it, so as a workaround we proxy for national banks with banks that contain the string "nat", which captures "National," "Natl.", etc.

[21] We query three LLMs: `gpt-5-mini`, `gemini-3-flash-preview`, and `deepseek-v4-pro`. If all three models agree on the episode type, we select the answer from the first one in this list. Otherwise, we apply a majority vote with priority-based tie breaking in the order listed above.

[22] Due to cost and context-window limitations of current LLMs, we only include up to 25 articles for each bank episode queried. For episodes that exceed this count, we try to preserve a representative sample of articles discussing different events (runs, suspensions, etc.) while selecting for high-quality articles within each sample.

- *Run-suspension-failure.* A bank run takes place. The bank suspends all payments. The bank remains closed permanently and has a receiver assigned (or similar resolution process for some non-national bank types).

Further, we also consider episodes without a run but where the bank suspends temporarily or permanently:

- *Suspension-failure*. A bank suspends. There is no recorded bank run. The bank remains closed permanently and has either a receiver assigned or is taken over by another bank.

- *Suspension-reopening*. A bank suspends, but there is no run on this specific bank. This can, for example, occur as part of a clearinghouse decision to suspend following runs on other banks. However, the bank reopens.

Note that by construction a bank must suspend before or at failure. Thus, any episode with failure will also involve a suspension. For our analysis using national banks, we rely on information on whether a bank failed directly from the OCC Annual Report to Congress digitized in Correia, Luck and Verner (2025).

Measurement Issues We note three measurement issues that are important to keep in mind when using information on bank distress recorded in newspapers. First, our database employs a binary classification of whether a run occurs according to newspapers. However, the severity of runs can vary, as we will show using data on bank deposits.

Second, newspapers may not always report runs. This concern applies especially to smaller banks. As with any narrative classification, there is always a concern that an event is not discussed in the narrative accounts under consideration. There can be episodes of "quiet" distress. Nevertheless, we do find that newspapers contain a massive amount of valuable information when validating our measure against existing narrative chronologies of banking crises or existing information on bank failures for national banks, as we show in the next section. Moreover, the typical run on a national bank in our dataset is discussed by 9 separate underlying newspaper articles, indicating that runs were generally newsworthy and widely reported.[23]

Third, during crises, newspapers sometimes speak more generally about several banks in an area being subject to runs or suspension, without explicitly listing all banks. Our baseline classification of a bank run requires a newspaper article to explicitly mention

[23]Table C.2 provides statistics on the number of articles used to construct each episode. The episode with the most newspaper articles is the run, suspension, and reopening of the Knickerbocker Trust Company during the Panic of 1907.

that a specific bank was run. However, when a newspaper clearly and explicitly reports that all banks in a city were subject to a run (or all banks of a given type, such as all savings banks), we augment our classification to include runs on these banks. We conduct robustness tests when including these "all banks were run" cases and find similar results.

3.2 Additional Data Sources

We combine the list of bank distress episodes with bank financial data for national banks. We use balance sheets of national banks from 1863-1941 from Correia, Luck and Verner (2025), based on national bank call reports submitted in the OCC's Annual Report to Congress.[24] These data contain detailed balance sheet line items, including total assets, loans, securities, deposits, equity capital, and noncore (non-deposit and non-equity) funding.

For national banks, we use the precise information on closures of national banks (receivership, mergers, assignment, and voluntary liquidations) from van Belkum (1968), Huntoon (2023), and Correia, Luck and Verner (2025). We define a national bank as failed when the OCC appoints a receiver. This definition of failure includes banks that eventually exit receivership, restore solvency, and continue operating, as well as banks that exit receivership and wind down their operations in an orderly voluntary liquidation that imposes no losses to creditors. However, this definition explicitly does not include temporary suspensions. To measure the extent of the deposit outflow in a bank failure, we further use deposits at suspension based on data on OCC receiverships digitized by Correia, Luck and Verner (2025). Given that information on bank fundamentals and closures is consistently available only for national banks, most of our analysis focuses on the sample of national banks.

To capture local defaults of non-financial firms that can trigger bank asset losses, we create a novel state-level dataset of quarterly non-financial business failures covering 1886–1935 by digitizing various issues of *Bradstreet's*. To obtain a business failure rate, we scale non-financial business failures by the number of manufacturing establishments from the census. To proxy real economic activity at the city level, we also create an index of manufacturing activity from *Bradstreet's* Trade at a Glance tables (1917-33) and descriptions of Business Conditions (1933-35). The index is consistently available every week for 30 cities from 1917 through 1933 and every month from 1933 to 1935. Appendix D provides additional details and shows that the index is highly correlated with aggregate

[24]The data in Correia, Luck and Verner (2025) build on and extend the data in Carlson, Correia and Luck (2022) and are digitized using the OCR techniques developed in Correia and Luck (2023).

industrial production ($R^2 = 49\%$). Finally, we use information on banking crisis years and the month of banking panics from Baron, Verner and Xiong (2021), as well as information on regional banking panics from Wicker (1996) and Jalil (2015).

4 Bank Runs in the US, 1863-1934

This section provides an initial look at the incidence of bank runs, suspensions, and failures in the US from 1863 to 1934. In addition to providing basic descriptive statistics, we also validate the quality of our newly constructed database, a key step to ensure the reliability of our findings.

4.1 Descriptive Statistics on Bank Distress Episodes

Panel (a) of Table 1 reports the raw counts of bank runs, failures and suspensions identified in our new database for all bank types. Figure 1 shows a Venn diagram of episodes with a run, suspension, and/or failure. As noted above, each observation represents an episode of bank distress for a particular bank, with the same bank possibly being subject to distress at different points in time over the sample period.

The majority of runs do not involve bank failure, and the vast majority of failures do not involve runs. During the 1863–1934 period, our dataset contains 3,984 bank run episodes. The first key insight is that the majority of runs do not involve failure: There are 2,074 runs without failure and 1,910 runs with failure. We also find 13,772 episodes involving a suspension, and 10,341 episodes involving a failure.[25] A second key insight is thus that runs are substantially less common than bank suspensions and failures, and most bank failures do not involve runs, as illustrated in the Venn diagram (Figure 1).

Panel (b) in Table 1 shows the unconditional probability of bank distress episodes in our 1863-1934 panel of national banks. To calculate probabilities of bank distress, we focus on national banks, where we have a precise measure of the total number of banks for the entire sample. Of the 3,984 bank run episodes, 1,280 are runs on national banks. For national banks, we also have clear records that determine whether a bank failed, as the OCC recorded all national bank receiverships. Such precise records are not available for state banks.[26] A national bank's annual unconditional probability of failing between

[25]As noted above, a bank always suspends before or at failure. Hence, by definition, there are no failures without suspensions.

[26]There is no census on the universe of non-national banks. Moreover, there is no systematic information on failures of state banks or trusts. Therefore, much of our analysis relies on the merged sample of national banks.

Table 1: ***Bank Runs, Failures, and Suspensions in Newspapers, 1863-1934***

Panel A: Total number of events, all bank types

Sample	Bank run				Failure	Suspension
	All	Run w/o suspension	Run with sus. + reopen.	Run with failure		
All	3,984	1,435	639	1,910	10,341	13,772
No crisis	2,130	823	232	1,075	6,198	7,245
Banking crisis	1,854	612	407	835	4,143	6,527
1863-1913 (NB Era)	2,042	991	341	710	3,717	4,932
1914-1928 (Early Fed)	797	230	87	480	2,655	3,082
1929-March 6, 1933	1,132	208	210	714	3,367	4,825
After March 6, 1933	13	6	1	6	602	933

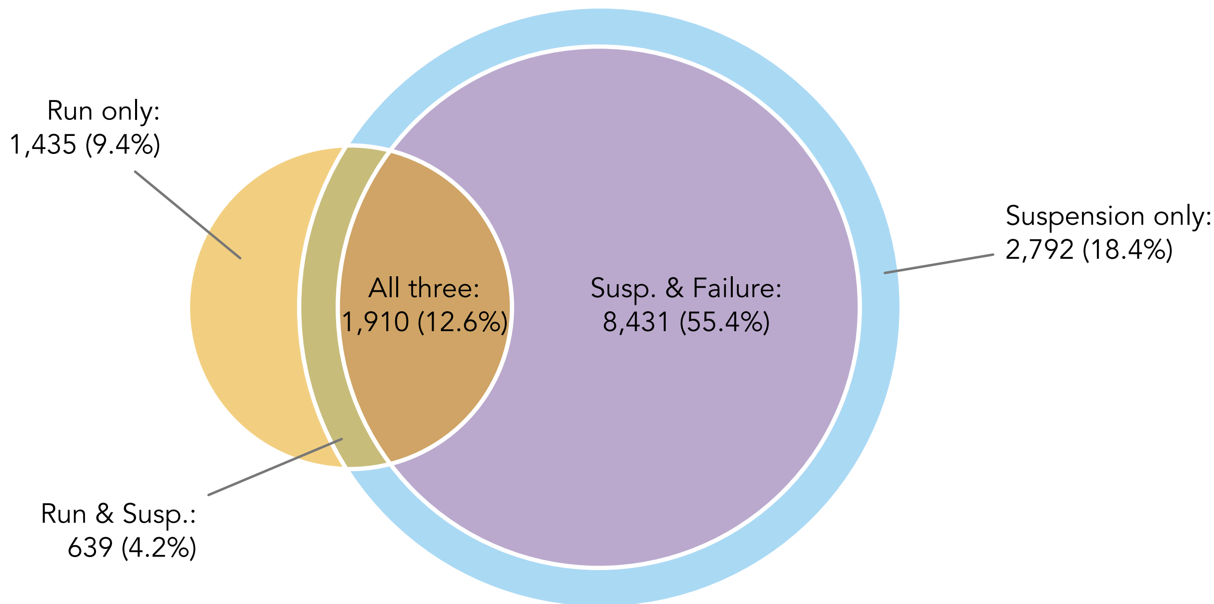
Panel B: Number of events as a share of total number of banks (in %), national banks

Sample	Bank run				Failure	Suspension
	All	Run w/o suspension	Run with sus. + reopen.	Run with failure		
All	0.39	0.15	0.06	0.18	0.89	1.17
No crisis	0.23	0.09	0.02	0.11	0.49	0.55
Banking crisis	1.42	0.49	0.29	0.65	3.45	5.07
1863-1913 (NB Era)	0.39	0.18	0.08	0.13	0.32	0.53
1914-1928 (Early Fed)	0.21	0.07	0.01	0.12	0.61	0.64
1929-March 6, 1933	1.03	0.24	0.12	0.68	3.68	4.95
After March 6, 1933	0.03	0.01	0.00	0.03	3.50	4.09

Notes: Panel A reports the counts of all runs, suspensions, and failures we identify in newspapers from 1863 to 1934. The sample covers bank distress episodes for all bank types: national banks, state banks, trust and private banks. Panel B reports the average annual rate of runs, suspensions, and failures for national banks, defined as the number of distress events indicated in the table header divided by the number of national banks in the previous year. For this sample, a bank failure is defined as an OCC receivership. The number of national banks and OCC receiverships per year comes from Correia, Luck and Verner (2025). Both panels distinguish between runs without suspension ("run only"), runs with suspension and reopening, and runs with failure. We also split the sample by different eras (National Banking Era, Early Federal Reserve, and Great Depression before and after the 1933 banking holiday) and whether a year involved a major banking crisis as identified in Baron, Verner and Xiong (2021).

1864 and 1934 is 0.89%, while the unconditional probability of being subject to a run is 0.39%. Again, the annual probability of a run without failure is 0.20%, so more than half of runs on national banks are not associated with failure. About 28% of runs without failure involve temporary suspension.

Figure 1: ***Bank Run, Suspension, and Failure Episodes***



Notes: The sample is all bank distress episodes involving a run, suspension, and/or bank failure over the period 1863-1934 for all bank types. By definition, failure requires suspension, so there are no cases with failure but without suspension.

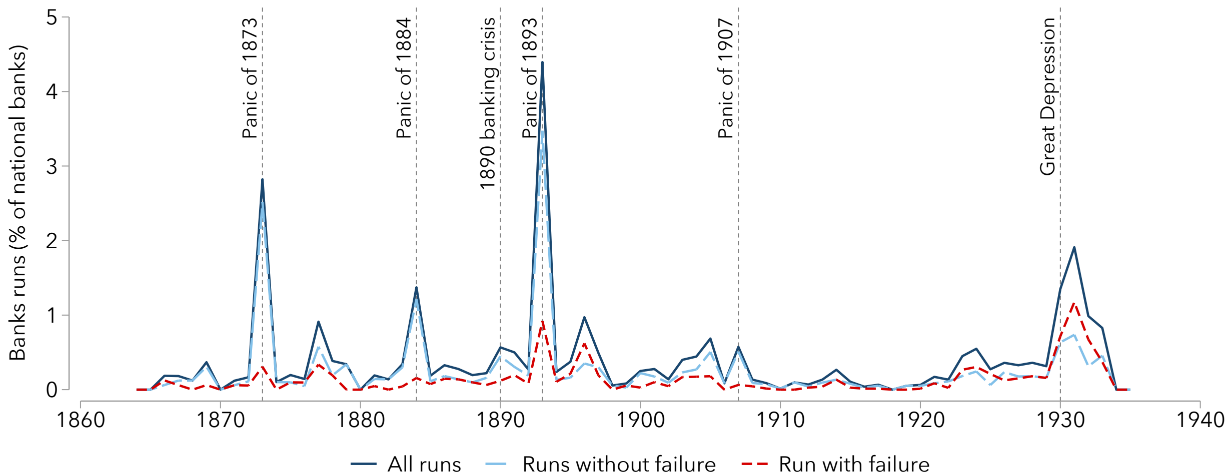
4.2 Bank Runs and Narrative Banking Crises

Figure 2 presents the rate of runs in the time series by plotting the number of runs as a fraction of the total number of banks in each year.[27] We again focus on national banks so that we can calculate rates relative to the total number of banks. The rate of runs spikes during the major crisis years: 1873, 1884, 1893, 1907 (to a lesser extent), and the Great Depression. This incidence of runs confirms prominent narratives of crises in the pre-FDIC era (e.g., Calomiris and Gorton, 1991; Wicker, 1996, 2006). The highest rate of runs occurs in 1873, 1893, and 1930-33, which involved the most severe crises in the commercial banking system. The spike in runs is smaller for the milder crises of 1884 and 1890, as well as for 1907, where distress was more pronounced among non-banks (trusts) and in the money market. Appendix E provides case studies of the major runs and failures

[27] Appendix Figure A.1 plots the fraction of national banks subject to a distress episode involving a run, suspension, or failure. Runs and suspensions spike during all major banking crisis years. Failures spike during some of the crisis years, such as 1893 and the Great Depression. We note that we "only" identify around 700 suspensions after the Banking Holiday in 1933 when all banks suspended (see, e.g., Jaremski, Richardson and Vossmeyer, 2023). This is explained by the fact that newspapers may not have seen a need to report individual bank suspensions right around the time the entire banking system suspended.

in these crises, illustrating that our database captures the well-known bank-level events discussed in prominent narratives of U.S. banking crises (e.g., Sprague, 1910; Friedman and Schwartz, 1963; Wicker, 1996, 2006; Rockoff, 2021; Conti-Brown and Vanatta, 2025).

Figure 2: ***Bank Runs With and Without Failure, National Banks, 1863-1934***



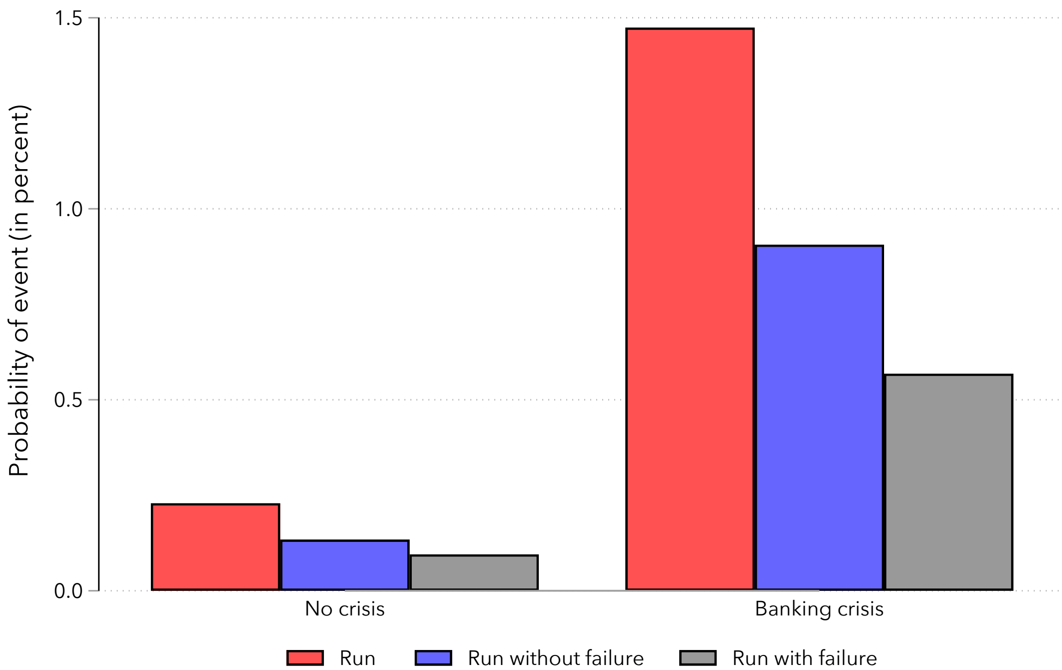
Notes: This figure plots the ratio of the number of national banks subject to a run over the total number of national banks in each year. The number of national banks each year is from Correia, Luck and Verner (2025). The figure also separates runs into runs without failure and runs with failure.

Figure 2 also distinguishes between bank runs with and without failure. Runs without failure are especially pronounced during major banking crises, especially the "panics" of the national banking era. The Panics of 1873 and 1893 featured a particularly large number of runs without failure (but with temporary suspensions), as noted by previous work (e.g., Carlson, 2005). Runs with failure spiked during the Panic of 1893 and the Great Depression. Interestingly, most failures do not involve discussions of runs. Failures without runs were elevated during the agricultural downturns of the 1890s and 1920s; see Appendix Figure A.1.

In the time series, runs are much more likely during major banking crisis years. Figure 3 plots the rate of runs during and outside of banking crisis years and confirms this finding. Crisis years are based on the narrative chronology by Baron, Verner and Xiong (2021).[28] The probability of a run increases by roughly a factor of six during banking crisis years compared to other years. The probability of a national bank being subject to a run is around 0.2% outside of banking crisis years but increases to more than 1.4% during a crisis; see also panel (b) of Table 1. The probability of a failure increases from 0.49% to 3.5%, and the probability of a suspension from 0.55% to 5.1%.

[28]Baron, Verner and Xiong (2021) define panics as episodes of severe and sudden withdrawals of funding by bank creditors from a significant part of the banking system.

Figure 3: ***Probability of Bank Runs With and Without Failure for National Banks During and Outside of Banking Crises***



Notes: The figure plots the probability of bank runs, runs without failure, and runs with failure during and outside of years identified as banking crises in Baron, Verner and Xiong (2021). The sample is all national banks during 1863-1934. Figure A.3 plots the probability of runs, suspensions, and failures during and outside of crises.

Table 2 confirms that bank runs and other distress events recorded in newspapers are substantially more likely in years and U.S. states that have been classified as having a banking crisis/panic by narrative chronologies of crises in a regression framework. Specifically, we test whether our list of bank distress events lines up with the state-level narrative banking panic dates from Jalil (2015) (pre-1930) and Wicker (1996) (1930-33), and with the aggregate banking crisis dates from Baron, Verner and Xiong (2021).[29] With this narrative information, we estimate the following regression at the state-month level:

$$\begin{aligned} \text{Number of bank distress events}_{st} = \beta_0 &+ \beta_1 \times \text{Major banking crisis}_t \\ &+ \beta_2 \times \text{Major panic month}_t \\ &+ \beta_3 \times \text{Regional banking panic}_{st} + \epsilon_{st}. \end{aligned} \tag{1}$$

[29] Jalil (2015) provides information on state-level banking panics. Jalil (2015) also reports city-level panics, which we aggregate to the state level. Wicker (1996) provides a detailed descriptive account of the regional panics during the Great Depression, which we hand-code from his text. Baron, Verner and Xiong (2021), as noted above, provide years of the onset of banking crises and the starting month of major banking panics.

The dependent variable is the number of runs, suspensions, or failures at the state-month level. Major banking crisis$_t$ is a dummy that takes a value of one in years of major banking crises, and Major panic month$_t$ is a dummy that is one in months of major banking panics. Regional banking panic$_{st}$ is a dummy that takes the value one if a state is subject to a banking panic in a given month. We compute robust standard errors clustered by state.

Table 2: ***Bank Runs, Suspensions, and Failures Correlate with Narrative Banking Crisis Classifications***

Dependent variable	Run	Failure	Suspension
	(1)	(2)	(3)
Major banking crisis year	0.20***	0.38***	0.68***
	(0.030)	(0.074)	(0.11)
Major banking panic month	0.89**	-0.34***	0.90**
	(0.43)	(0.095)	(0.45)
Regional banking panic	2.17***	2.34***	5.37***
	(0.47)	(0.43)	(0.95)
N	41609	41609	41609
Mean dep. var.	0.10	0.19	0.28

Notes: This table reports estimates of Equation (1) using a monthly state-level panel with the number of distress events as the outcome variable. Major banking crises and panic months are taken from Baron, Verner and Xiong (2021). This chronology captures the start year of banking crises; therefore, for the Great Depression, we also include 1931–1933 as crisis years. Regional banking crises are from Jalil (2015) (1884-1929) and Wicker (1996) (1930-1933). Table A.1 presents estimates separately including a dummy for each regional crisis. The sample period is 1863-1934. We compute robust standard errors clustered by state. *,**, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Our database of bank distress events correlates strongly with existing narrative chronologies of banking crises in U.S. history. On average, there are 0.1 bank run events per state-month. During years with a banking crisis, the number of runs in a state-month triples, increasing by 0.2 on average.

Moreover, bank run spikes also line up with the month of banking panics from narrative accounts. During panic months, the number of runs increases by another 0.9. This result is also confirmed by Figure A.2 in the Appendix, which plots the rate of bank runs at the monthly frequency. With vertical lines, we plot the monthly start date of banking panics according to Baron, Verner and Xiong (2021). During the National Banking Era, newspapers also pick up several additional borderline cases of banking panics, such as 1896. For the Great Depression, we capture the various waves of runs discussed in narrative accounts. In particular, we capture the rise in runs in late 1930, the rise in runs from March to August 1931, the spike in runs after the U.K. leaves gold in September 1931, and another rise from the end of 1932 to the banking holiday in March 1933, consistent with the aggregate narrative of Friedman and Schwartz (1963). We also capture a small spike in runs in early 1930 and during the Chicago panic in the summer of 1932 (Calomiris and Mason, 1997). Further, while bank runs were more likely before the banking holiday of 1933, failures and suspensions were mostly reported during or after the banking holiday, in line with Jaremski, Richardson and Vossmeyer (2023).

Table 2 further shows that during months of regional banking panics identified by Jalil (2015) and Wicker (1996), the number of runs increases by 2.2 in a state. Bank suspensions and failures display a similar pattern. Further, in Table A.1, we include dummies for individual regional crises identified by Jalil (2015) and Wicker (1996) in the regression. Most individual crises see considerable increases in the number of runs and failures. For instance, the number of runs increases by 9 during the Florida "Fruit Fly Panic" in July 1929 (Carlson, Mitchener and Richardson, 2011) and by 15 in the June 1932 Chicago panic (Calomiris and Mason, 1997).

Taken together, this evidence indicates that our database of runs lines up quantitatively with existing qualitative narrative chronologies of banking crises at the aggregate and regional level. The detailed case studies of bank-level events during major banking crises in Appendix E support this statistical evidence. These case studies, with links to original news articles, show that we capture the famous bank runs and failures of all major crises in our sample period.

4.3 Data Quality Checks

Beyond showing that our bank distress events dataset lines up with aggregate and regional banking crisis chronologies, we provide several additional formal validation exercises.

Cross-Validation Exercises To quantify the accuracy of our data construction pipeline, we conduct four formal cross-validation exercises. These exercises are detailed in Appendix C.4. Overall, the match rate for national bank events is about 95% across the four exercises.

First, we build a ground truth dataset of 287 runs, suspensions, and reopening events based on reading Wicker (1996, 2006). These two studies provide detailed accounts of bank-specific events from historical newspapers. For national banks, the key sample for most of our analysis, our dataset matches 94.1% of runs and 94.9% of all events in this ground truth. For all banks, including state and private banks, we match 85.0% of runs. Importantly, relative to the descriptive account in Wicker (1996, 2006), we identify nearly 40 times more bank runs, and we can match these runs to bank fundamentals.

Second, we match all but one event from a list of 19 prominent runs and failures discussed in Rockoff (2021), yielding a match rate of 94.7%. We match all national bank events in this list. The only missing event is the failure of a trust that is not in our bank census.

Third, we match 93.5% of suspensions from a list of 123 national bank suspensions published by the trade journal *Bradstreet's* following the Panic of 1893.

Finally, we hired eight undergraduate students for one month to search newspapers.com for runs across our full sample period. This provided us with a sample of 271 bank runs from 1863 through 1934 that we manually verified. Compared to the previous three ground truth datasets, this last dataset provides a sample of somewhat more random runs discussed in newspapers, as it is not based on prominent runs or events during major crises. Our dataset matches 94.3% of the runs on national banks in this ground truth. Extrapolating linearly, a heroic assumption, constructing our new distress events database through hand collection would require roughly 9,400 human hours, in addition to the untold cost of dissuading a group of promising research assistants from becoming economists.

We find that a bank event is not matched for one of two main reasons: either the bank is not in our bank census (not a concern for national banks) or we do not have a news article on the particular event in our corpus. Reassuringly for our LLM-based approach, we found that false positives and false negatives caused by LLM errors are very rare. Conditional on high-quality news article inputs, the ensemble of LLMs is highly accurate in classifying runs and other distress events.

Additional Data Quality Checks We provide four additional data quality checks. First, we correlate the true failure rate for national banks based on official OCC receiverships with the failure rate according to our classification from newspapers. The aim of this exercise is to assess how informative newspapers are about bank distress events whose occurrence we can verify with certainty. We construct an alternative failure rate from our bank distress episodes database by dividing the number of national bank failures in newspapers by the total number of national banks. Figure A.4a shows that the overall correlation between the true failure rate and the failure rate based on newspapers in annual data is good, but not perfect, as one might expect. This indicates that newspapers are informative about important bank distress events.

Second, we study whether deposit outflows in failing banks are larger for bank failures involving a run. Figure A.4b plots the distribution of actual deposit outflows in national bank failures for failures with and without a bank run according to our bank distress episodes database. Deposit outflows are measured as the growth in deposits from the last call report before suspension to the time of suspension. The figure shows that failures with narrative evidence of a run in newspapers have larger deposit outflows before failure. On average, failures with narrative evidence of a run exhibit a 5 percentage points larger deposit outflow before failure. This finding provides evidence that narrative accounts of runs in newspapers indeed capture many failures with large deposit outflows.[30]

The previous exercise focuses on deposit outflows conditional on bank failure. As a third exercise, Figure A.4c plots the probability of a bank distress episode as a function of annual deposit growth, conditional on survival. Bank runs and suspensions that do not result in failure are all significantly more common for banks in the bottom 1% of the deposit growth distribution. Thus, our classification of runs correlates with statistical evidence of deposit outflows.

Finally, Figure A.4d plots the probability of a run or failure reported in newspapers as well as the actual annual rate of bank failures by bank size, measured using total assets. It reveals that our bank distress events database from newspaper accounts is more likely to report events for larger banks, in line with events for larger banks being more newsworthy. Thus, the incidence of bank runs is more likely to be under-reported for smaller banks. To address this potential shortcoming of our data, we show below that our main findings are robust to restricting the sample to large banks.

5 Determinants of Bank Runs

What are the characteristics of banks subject to runs? Which shocks trigger runs? Are the determinants of runs the same as the determinants of bank failure? Guided by the predictions of the theories discussed in Section 2, this section presents evidence on the determinants of bank runs.

5.1 Empirical Framework

We start by assessing the likelihood of a bank run as a function of bank-specific and broader economic conditions. As discussed in Section 2, theories of bank runs have differ-

[30]Note that the deposit outflow from the last call report to failure can capture a steady deposit drain over a period of up to a year. Therefore, there are failures with larger deposit outflows that do not necessarily involve a run based on our conceptual definition of a run.

ent predictions for what can explain the incidence of runs. Theories of non-fundamental runs (e.g., Diamond and Dybvig, 1983) suggest that even healthy banks can be subject to a run, thus reducing the explanatory power of bank fundamentals. In contrast, theories of fundamental-based panic runs (e.g., Goldstein and Pauzner, 2005) suggest that runs should predominantly happen in banks with weak fundamentals. Finally, theories of information-based runs (e.g., Gorton, 1988) suggest that runs are more likely after adverse public signals about the state of the banking sector or the macroeconomy.

Motivated by these predictions, we examine the ability of various sets of variables to capture an elevated risk of runs and compare the determinants of bank runs to those of bank failure. Our goal is to describe the likelihood of runs as a function of bank-specific fundamentals and public signals that may trigger runs. To do this, we estimate predictive regressions of the following form:

$$\text{Run}_{bt+1} = \beta_0 + X_{bt}^{Bank}\beta + X_t^{Macro}\gamma + X_{bt}^{Local}\delta + \epsilon_{bt+1}, \tag{2}$$

where Run_{bt+1} is an indicator variable for whether bank b is subject to a bank run within the next year. In addition to examining bank runs, we also present results for several other outcomes: bank runs that do not result in failure, runs that result in failure, and all failures. The model allows us to test whether runs are predictable, and whether factors that imply a higher probability of a run are similar to those indicating a higher risk of failure. We use the sample of all national banks from 1863 to 1934, for which we have information on both bank runs and bank-level fundamentals captured in bank balance sheets.

On the right-hand side, we include a range of fundamental bank balance-sheet metrics in X_{bt}^{Bank} to proxy for bank health and risk-taking. These measures have been shown to capture the risk of bank failures in prior work (e.g., White, 1984; Calomiris and Mason, 1997; Correia, Luck and Verner, 2025). First, we proxy for bank solvency with the surplus-to-equity ratio. Surplus profit captures past retained earnings and is thus a measure of past bank profitability and capitalization. Second, we include a measure of noncore funding. Noncore funding is defined as the ratio of non-deposit and non-equity funding to total assets. This measure includes wholesale funding, such as bills payable and rediscounts. Noncore funding tends to be more expensive and risk-sensitive than deposit funding. Existing work shows that reliance on noncore funding was often a signal of troubled banks suffering losses (e.g., White, 1984; Calomiris and Mason, 1997). Therefore, we note that, while noncore funding captures bank funding structure, it is both a signal of weak solvency and of potential liquidity risk. We also interact the solvency and noncore funding measures, as banks with low solvency and high reliance on noncore funding might be especially likely to experience future distress. Third, we include the liquid assets ratio as a proxy for both a bank's available liquid funds to accommodate withdrawals and the riskiness of a bank's investment mix. Fourth, we include deposits-to-assets to capture the reliance on deposit funding. Finally, we include recent asset growth. Troubled banks might contract shortly before failure, so asset growth may contain information about the future likelihood of failure.

We also examine whether adverse public signals that are salient to depositors imply an increase in the likelihood of a run. In X_t^{Macro}, we include aggregate information that is not specific to bank b that might lead depositors to revise their probability that banks are troubled. Specifically, we include the following time series variables: annual total return on the stock market, real GDP growth, the aggregate national bank failure rate, the aggregate rate of runs on national banks, and the aggregate non-financial business failure rate. In X_{bt}^{Local}, we include local public signals: an indicator for whether there is a run on another bank in the same city and the state-level non-financial business failure rate. All variables are measured as of year t. For example, the stock market return used to predict a run in year $t+1$ is measured as the return from December $t-1$ to December of t.

In addition to the estimated coefficients, we are interested in the extent to which bank-specific fundamentals and economic conditions can explain bank runs and other bank distress episodes. To assess the explanatory power, we calculate the in-sample area under the receiver-operating characteristic curve (AUC) for various predictive models. The receiver-operating characteristic (ROC) curve traces the true-positive rate against the false-positive rate across classification thresholds. The AUC is the integral under the ROC curve. An AUC of 1 indicates perfect classification performance, while an AUC of 0.5 indicates an uninformative predictor.

5.2 Determinants of Runs

Table 3 presents the results from predicting bank runs as a function of bank fundamentals, macro conditions, and local economic conditions. Column 1 uses only bank fundamentals. It shows that banks with low solvency (proxied by surplus-to-equity) and a high reliance on noncore funding are more likely to experience a run. The interaction between the solvency and noncore funding measures is also predictive of runs. Further, banks with a low share of liquid assets are more likely to be subject to a run. A higher deposits-to-assets ratio, indicating higher leverage, is also positively related to runs. These patterns are consistent with the notion that runs are more likely in banks that are closer to insolvency and less liquid, consistent with theories of fundamental-based runs where the threshold for whether a run occurs is a function of bank leverage and illiquidity.

Table 3: ***Predictability of Bank Runs Based on Bank Fundamentals and Economic Conditions***

Dependent variable	Run in $t+1$			
	(1)	(2)	(3)	(4)
Bank fundamentals:				
Surplus / Equity	-0.30**			-0.51***
	(0.13)			(0.19)
Noncore funding / Assets	11.06***			10.23***
	(1.78)			(1.83)
Surplus × Noncore funding	-11.05***			-11.75***
	(2.60)			(2.35)
Liquid Assets / Assets	-0.65***			-0.67*
	(0.24)			(0.34)
Deposits / Assets	0.74*			0.67*
	(0.40)			(0.34)
Asset Growth (3 years)	-0.15			-0.05
	(0.14)			(0.09)
Macro conditions:				
NB failure rate		-0.79		-0.29
		(3.96)		(3.73)
NB run rate		5.40		-5.30
		(17.17)		(14.63)
Business failure rate		21.34*		19.25
		(11.78)		(13.53)
Real GDP growth		0.93		0.33
		(0.89)		(0.72)
Stock market return		-0.68**		-0.65***
		(0.26)		(0.22)
Local conditions:				
Run on other bank in same city			3.47***	3.45***
			(0.95)	(0.95)
Local business failure rate			15.16***	1.95
			(3.60)	(6.28)
N	290437	277086	273481	243908
Mean dep. var	.39	.38	.38	.38

Notes: This table presents estimates of Equation (2). Driscoll and Kraay (1998) standard errors are reported in parentheses, with a bandwidth of three years to allow for residual correlation within and across banks in proximate years. *,**, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Column 2 in Table 3 focuses on the information content of adverse public signals about the economy and the banking system that are not bank-specific. We find that runs are more likely following years with low stock returns. They are also more likely when business failures are high.

Column 3 focuses on the predictive content of local economic conditions. Runs are significantly more likely following a run on another bank in the same city. The likelihood of a run is also elevated when the state-level business failure rate is higher. These findings are consistent with information-based runs, whereby agents observing adverse news about the economy or banking system revise their probability that other local banks are troubled, increasing the likelihood of runs on other banks. It can also be explained by deteriorating local economic conditions weakening many local banks.

5.3 Predictability of Runs With and Without Failure

Table 4: ***AUC Metric for Predictability of Bank Runs and Bank Failures***

Prediction horizon h	1 year					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Bank run						
AUC	0.692	0.817	0.749	0.788	0.799	0.840
Panel B: Bank run without failure						
AUC	0.617	0.798	0.687	0.734	0.755	0.818
Panel C: Bank run with failure						
AUC	0.838	0.878	0.865	0.878	0.883	0.895
Panel D: Bank failure						
AUC	0.874	0.903	0.897	0.892	0.904	0.911
Specification details						
Bank fundamentals	✓	✓	✓	✓	✓	✓
Macro conditions			✓		✓	
Local conditions				✓	✓	✓
Year FE		✓				✓

Notes: This table reports the area under the receiver operating characteristic curve (AUC) across different outcome variables (rows) and specifications (columns). The corresponding regression coefficients underlying the models are reported in Table A.2 (for Panel A), Table A.3 (for Panel B), Table A.4 (for Panel C), and Table A.5 (for Panel D).

Table 4 studies how well the three sets of variables can classify bank runs and other bank distress episodes based on the AUC metric. Panel A focuses on the predictability of bank runs. Panel A, column 1 shows that bank runs are only moderately predictable based on bank fundamentals, with an AUC of 0.69.

Table 4 column 2 includes year fixed effects in the predictive regression. This specification asks what is the highest AUC that can be obtained from annual time series variation. Including year fixed effects substantially increases the AUC for predicting runs to 0.82. Thus, time series variation is an important source of variation in runs. This is consistent with runs clustering in particular "panic" years.

In columns 3-5, we attempt to "interpret" the year fixed effects by including the proxies of economic conditions. Adding information on local conditions substantially increases the AUC for bank runs, moving the AUC closer to the specification with year fixed effects. The AUC of 0.80 in column 5 indicates that runs are substantially, though not extremely strongly, predictable based on bank-level and aggregate information.

We next compare the predictability of bank runs with and without failure (Table 4 panels B and C). We also compare the predictability of runs to the predictability of bank failures (panel D). Column 1 in Table 4 reveals that bank fundamentals are much more strongly related to bank runs *with* failure compared to bank runs without failure. Runs without failure are weakly predictable based on bank fundamentals, with an AUC of only 0.62. In contrast, bank failures are tightly linked to bank fundamentals with an AUC of 0.87 (panel D).

Column (2) shows that including year fixed effects substantially increases the AUC for predicting runs *without* failure. The relative increase is more modest for predicting runs *with* failure. This suggests that time series variation is especially important for understanding runs *without* failure, while bank-specific fundamentals are more important for classifying runs with failure. As a result, the inclusion of variables capturing macro and local conditions leads to a substantial increase in the AUC for bank runs without failure, from 0.62 to 0.76. However, it leads to a more moderate improvement for predicting runs with failure, which are already well-captured by bank-specific fundamentals.

Overall, the evidence indicates that, to understand bank runs, both bank fundamentals and adverse local and macro news play an important role. For runs without failure, adverse macro news is substantially more important than bank-specific information. This is in line with banking panics occurring after bad news about the economy or the banking system, leading to runs even on healthy banks. While balance sheet metrics contain substantial information for distinguishing strong and weak banks, confused depositors do not fully incorporate this information. On the other hand, bank fundamentals are much more important for understanding bank runs with failure, and bank failures more broadly. Bank failures, including those with runs, tend to occur in banks with the weakest balance sheets, consistent with the evidence in Correia, Luck and Verner (2025). Thus, fundamentals-based theories are promising for explaining bank runs and failures, while information-based theories are important for explaining runs without failure.

6 Bank Runs and Bank Failure

We next investigate the relation between bank runs and bank failures in more detail. Which runs lead to bank failure? Can a run lead to the failure of a healthy bank? Or do runs only lead to failures for banks with weak fundamentals?

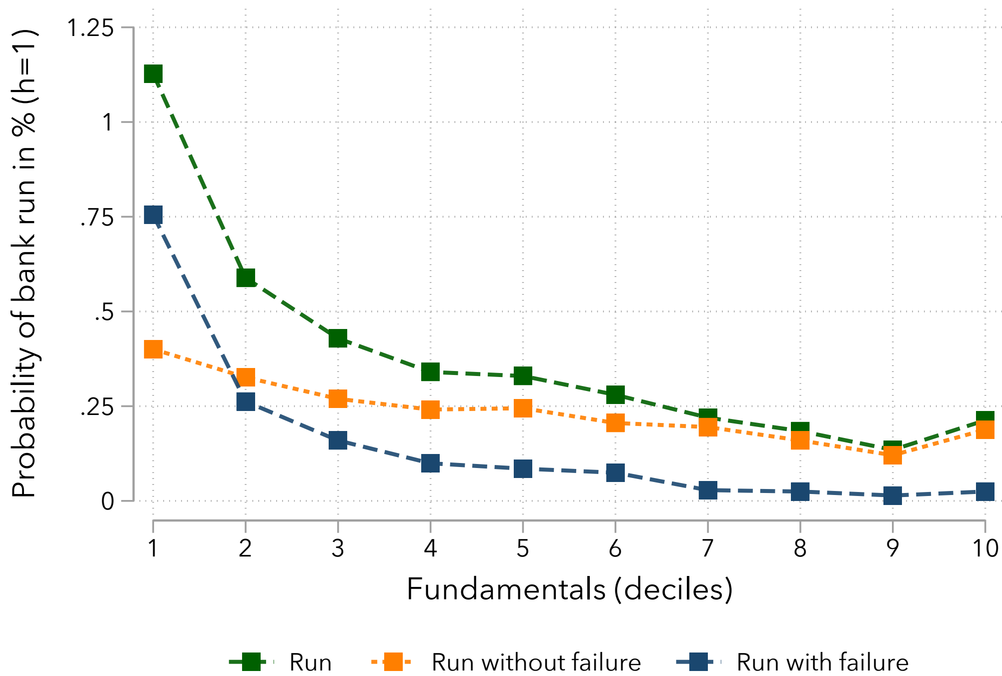
The previous section showed that runs are more likely in banks with weak fundamentals. Yet even strong banks can be subject to runs, especially in response to adverse public signals. In this section, we show that a run is much more likely to translate into failure for weak banks. Runs essentially never result in failure for banks with strong fundamentals. Moreover, failing banks display weak and deteriorating fundamentals, irrespective of whether they fail with or without a run.

6.1 Bank Runs and Failure across Weak and Strong Banks

Summarizing Fundamentals We start by constructing a simple bank-level measure of fundamentals, denoted Fundamentals$_{bt}$, based on the regression of bank failure on bank fundamentals (i.e., in column 1 of Table 4 with only bank fundamentals, reported in column 1 of Table A.5). The idea is to summarize fundamentals that are predictive of failure into one number, akin to an Altman Z-score (Altman, 1968). To avoid look-ahead bias, we estimate the regression iteratively using data up to $t-1$ to obtain predicted failure probabilities for time $t+1$. Fundamentals$_{bt}$ is then defined as the negative of the predicted value from the regression that predicts failure in year $t+1$ based on bank fundamentals, so a higher value indicates a bank with stronger fundamentals.

Fundamentals and Bank Runs With and Without Failure Figure 4 provides a simple visualization of the relationship between bank fundamentals and the likelihood of a run, a run with failure, and a run without failure. It shows the conditional probability of a bank run over the next year across deciles of our proxy for fundamentals. In line with the patterns documented in Table 3, runs are more likely in banks with weak fundamentals. A bank in the lowest decile of fundamentals has a roughly 110 basis point probability of being exposed to a run in a given year. The probability drops for stronger banks to below 25 basis points for banks with the strongest fundamentals. Thus, while much less likely, strong banks can be subject to runs. Moreover, we note that, while the probability of a run is higher for weak banks, both the level and the difference relative to strong banks

Figure 4: ***Probability of Bank Runs by Bank Fundamentals***



Notes: This figure plots the probability of a bank run, a bank run that does not result in failure, and a bank run that does result in failure across deciles of a proxy of bank-specific fundamentals (see text for definition).

are still relatively low in absolute terms, in line with the relatively modest predictability of runs based on fundamentals alone; see Table 4.

Figure 4 also plots the probability of runs with and without failure across the distribution of bank fundamentals. The figure reveals two distinctive patterns. Runs that do not result in failure are only weakly related to fundamentals. These runs are only slightly more likely to happen in weak banks than in strong banks.

Runs with failure, in contrast, are concentrated among banks with weak fundamentals. Their incidence is highest in the lowest decile and declines sharply for banks with stronger fundamentals. A bank that is above the 7th decile of the fundamentals distribution will very rarely be subject to a run and fail. Hence, a look at the relationship of runs, failures, and fundamentals in the raw data suggests that even though strong banks can be subject to a bank run, for a run to be associated with failure, a bank must have weak fundamentals.

Pass-through of Bank Runs to Bank Failure To more formally study the importance of bank fundamentals for whether a bank can withstand a run, we estimate the following specification for the pass-through of a bank run into failure:

$$\begin{aligned} \text{Failure}_{bt} = \beta_0 + \beta_1 \text{Run}_{bt} + \beta_2 \text{Fundamentals}_{bt-1} \\ + \beta_3 \text{Run}_{bt} \times \text{Fundamentals}_{bt-1} + \epsilon_{bt}, \end{aligned} \tag{3}$$

where Failure_{bt} is an indicator for whether bank b fails in year t, Run_{bt} is an indicator for whether the bank is exposed to a run in year t, and $\text{Fundamentals}_{bt-1}$ is our proxy for bank health measured in year $t-1$, defined above. This specification asks whether a run is associated with an increased likelihood of failure over the next year and how this varies across banks with weak and strong fundamentals.

Table 5: ***Pass-Through of Runs to Failure: The Role of Bank-Specific Fundamentals***

Dependent variable	Failure in t					
	(1)	(2)	(3)	(4)	(5)	(6)
Run	0.38***	0.25***	0.43***	0.40***	0.33***	0.29***
	(0.051)	(0.039)	(0.053)	(0.051)	(0.053)	(0.053)
Fundamentals		-3.40***				
		(0.82)				
Fundamentals × run		-12.7***				
		(1.82)				
Strong fundamentals			-0.0094**		-0.0054**	
			(0.0038)		(0.0025)	
Strong fundamentals × run			-0.32***		-0.27***	
			(0.049)		(0.048)	
Very strong fundamentals				-0.0072**		-0.0034**
				(0.0029)		(0.0017)
Very strong fundamentals × run				-0.30***		-0.23***
				(0.067)		(0.066)
Observations	272535	272535	272535	272535	73859	73859
Mean dep. var.	0.0084	0.0084	0.0084	0.0084	0.0050	0.0050
Sample					Large Banks	Large Banks

Notes: This table presents estimates of Equation (3). We define a bank to have "strong" fundamentals when $\text{Fundamentals}_{bt-1}$ is in the upper tercile of its historical distribution. "Very strong fundamentals" indicates when $\text{Fundamentals}_{bt-1}$ is in the highest decile. Driscoll and Kraay (1998) standard errors are reported in parentheses with a bandwidth of three years to allow for residual correlation within and across banks in proximate years. *,**, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 5 presents the results from estimating Equation (3) for the sample of national banks from 1863 to 1934. Column 1 shows that the probability of bank failure increases by 38 percentage points when a bank is subject to a bank run, roughly in line with Figure 1. This increase in the probability of failure is considerable given that the unconditional probability of failure is 0.84% in this sample. Thus, a bank subject to a run is more than

40 times more likely to fail than the average bank, and runs often result in failure. At the same time, a run is not a death sentence, and runs without failure are more common than runs with failure.

Column 2 in Table 5 further shows that banks with strong fundamentals have a significantly lower probability of failure, echoing the findings in Correia, Luck and Verner (2025). Moreover, in line with the evidence from Figure 4, the negative coefficient on $\text{Fundamentals}_{bt-1} \times \text{Run}_{bt}$ implies that strong fundamentals significantly reduce the probability of bank failure after a run.

In order to quantify how fundamentals matter for whether a bank fails after a run, we further interact the run indicator with an indicator for whether a bank's fundamentals are "strong." We define a bank as having "strong" fundamentals when $\text{Fundamentals}_{bt-1}$ is in the upper tercile of its historical distribution. Column 3 shows that banks in the two bottom terciles of the fundamentals distribution have a 43 pp higher chance of failing when subject to a run, but banks with strong fundamentals have a 32 pp lower probability of failure. Thus, having strong observable fundamentals substantially reduces the probability of failure when a bank is exposed to a run. Column 4 further shows that banks with very strong fundamentals, defined as being in the top decile of $\text{Fundamentals}_{bt-1}$, have a 30 pp lower probability of failure compared to all other banks, which have a 40% chance of failing in this specification. Thus, very strong banks rarely fail when subject to a run. While the fact that very strong banks can be subject to runs is in line with both theories of non-fundamental runs and information-based runs (Prediction 1), the fact that strong banks do not fail after a run is only predicted by theories of information-based bank runs (Prediction 2).

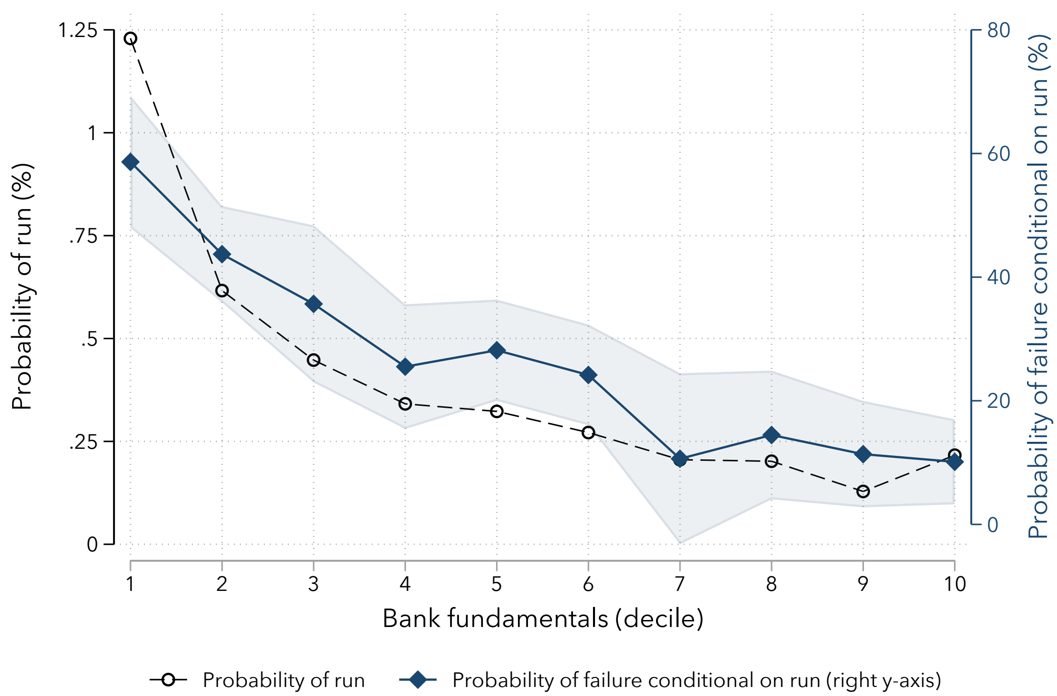
Figure 5 presents a visualization of the likelihood that a run leads to failure across the full distribution of bank fundamentals. For each decile d of $\text{Fundamentals}_{bt-1}$, we estimate

$$\text{Failure}_{bt} = \beta_0^d + \beta_1^d \text{Run}_{bt} + \epsilon_{bt}, \quad d = 1, ..., 10, \tag{4}$$

and plot the estimated coefficients $\{\hat{\beta}_1^d\}$. The figure shows that a run is associated with a 59 percentage point increase in failure probability for banks in the lowest decile of observable fundamentals. The increase in failure probability associated with a run falls nearly monotonically for banks with stronger fundamentals. For banks in the strongest decile of our proxy for fundamentals, the increase is only about 10 percentage points. Overall, this suggests that bank-specific fundamentals are central to understanding the pass-through of bank runs to failure.

Before proceeding, we note that these estimates provide upper bounds on the likelihood that a run causes a healthy bank to fail. The reason is that $\text{Fundamentals}_{b,t-1}$ may, in some cases, be an imprecise proxy of bank health. For instance, publicly available financial statements cannot capture episodes where a bank was insolvent but had not yet recognized losses (e.g., due to an unexpected shock, fraud, or hidden losses). In these cases, news about a bank's trouble that is not yet reflected in bank balance sheets would trigger a run and failure, but the run is a response to a signal of insolvency.[31]

Figure 5: ***Pass-Through of Runs to Failure across Deciles of Bank-Specific Fundamentals***



Notes: This figure plots the likelihood of a run (left y-axis) and the probability of failure conditional on a run (right y-axis) across deciles of bank fundamentals. The right y-axis plots the coefficients from estimating Equation (4) separately for each decile of the distribution of $\text{Fundamentals}_{bt-1}$. $\text{Fundamentals}_{bt-1}$ is a proxy of bank health (see text). The sample consists of all national banks from 1863 through 1934. The shaded area represents 95% confidence intervals based on Driscoll-Kraay standard errors with a bandwidth of three years to allow for residual correlation within and across banks in proximate years.

High Frequency Dynamics Figure 6 zooms in on the daily dynamics of the probability of failure following a run. The figure reports the cumulative probabilities of bank

[31] Correia, Luck and Verner (2025) document that failed national banks had both low recovery rates and low asset quality according to OCC bank examiners. That study argues this suggests that the vast majority of failed banks were fundamentally insolvent, unless one assumes large value destruction from receivership itself.

suspension and bank failure around a run for all national banks, estimated from local projections of the form:

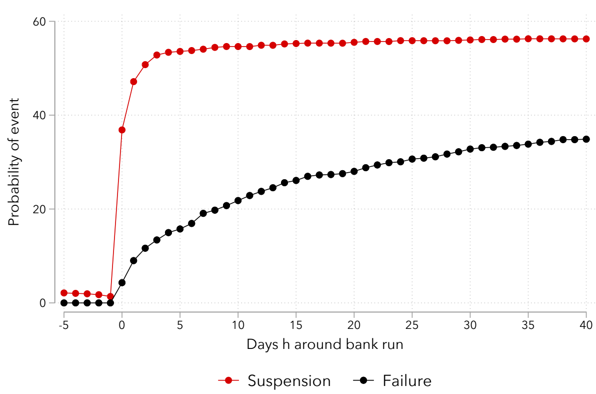
$$\text{Failure}_{bt+h} = \beta^h \text{Run}_{bt} + \epsilon_{bt+h}, \qquad h = -5, ..., 40. \tag{5}$$

Figure 6 panel (a) plots the cumulative probability of bank suspension and bank failure following a run. We find that the probability of bank failure rises gradually to about 35% in the 40 days following a run. Thus, essentially all failures following a run happen roughly within a month of the run. The dynamics of suspension display two differences compared to the dynamics of failure. First, suspension is more likely than failure, occurring in more than 50% of runs. Second, suspension typically occurs immediately, within the same day or within a few days of the run. Nearly 40 percent of runs involve suspension on the same day.

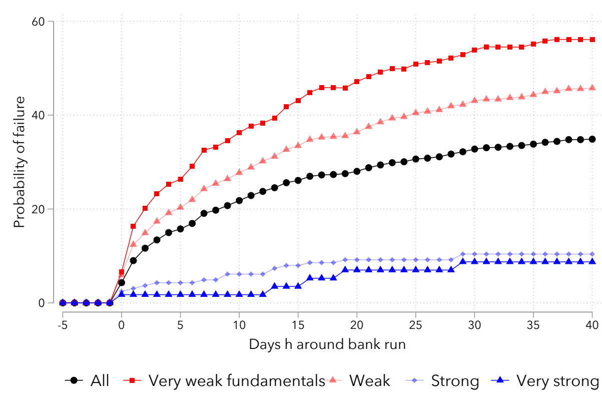
Figure 6 panel (b) further visualizes the cumulative probability of bank failure following a run across banks by their fundamentals. For each run occurring in year t, we split banks into the bottom decile (very weak), bottom tercile (weak), top tercile (strong), and top decile (very strong) of $\text{Fundamentals}_{bt-1}$, measured as of the most recent call report. In line with the evidence in Table 5, we find a striking difference in the likelihood of bank failure across weak and strong banks following a bank run. Very weak banks see a relatively quick spike in failures after runs and a cumulative probability of failure of about 56% within the 40 days following a run. In contrast, the probability of failure is about 9% for very strong banks.

Figure 6: ***Daily Dynamics of Failure and Suspension Probabilities around Bank Runs***

(a) *Suspensions and bank failures around bank runs*



(b) *Bank failure around runs by bank fundamentals*



Notes: Both panels plot coefficients from estimating Equation (5). In panel (a), we report results with failure and suspension as the outcome variables. In panel (b), we estimate the responses separately based on bank fundamentals, measured with $\text{Fundamentals}_{bt-1}$. Specifically, we define bank fundamentals categories as: "very weak" (bottom decile), "weak" (bottom tercile), "strong" (top tercile), and "very strong" (top decile).

Event Study of Fundamentals around Runs and Failures Appendix A.1 provides further illustration of the marked differences in bank fundamentals across banks subject to runs that fail and those that survive. We conduct simple event studies of bank fundamentals around runs without failure and runs with failure (see Figure A.5). Banks subject to runs that do not fail look similar to ordinary banks not subject to any form of distress. In contrast, banks that are run and fail have worse fundamentals five years before failure. Moreover, these banks have deteriorating profitability, rising non-performing loans, a declining deposit base, and an increasing reliance on expensive noncore funding in the three-to-five years before failure. This evidence reinforces the finding that poor fundamentals are key to determining whether a bank will fail. Runs that do not result in failure tend to occur in healthy banks, while runs that result in failure occur in observably weak banks.

In Figure A.5, we also compare banks that fail with a run to banks that fail without a narrative indication of a run. Banks that fail without a run are slightly worse across a range of fundamentals compared to banks that fail with a run, but the differences are modest. This suggests that runs may accelerate the failure of banks with deteriorating fundamentals, triggering the failure of marginally healthier banks. However, given that these banks are on very similar trajectories, many failures with runs were likely inevitable.

Robustness and Subsamples As noted above, bank distress episodes are more likely to be accurately reported in newspapers for the largest banks. Runs might be underreported for small banks, especially if they do not result in failure, possibly leading to an upward-biased estimate of the likelihood that a run results in failure for small banks. To address this concern, it is useful to verify that our findings hold in the sample of larger banks. Column 5 of Table 5 zooms in on banks in the top quartile of total assets. In the sample of large banks, we find that banks have a 33 pp higher chance of failing when subject to a run, but banks with strong fundamentals have a 27 pp lower probability of failure. These numbers are similar to the full sample estimates.

To ensure that the results are not sensitive to the construction of the summary fundamentals measure, in Table A.6, we estimate a variant of Equation (3) where we interact Run_{bt} with the underlying bank fundamental metrics, rather than using the summary $\text{Fundamentals}_{bt-1}$ measure. We find that the pass-through from runs to failure is lower for banks with higher surplus-to-equity, lower reliance on noncore funding, and a lower share of a proxy for nonperforming loans.

In Table A.7, we estimate the pass-through of runs to failure across subsamples. The pass-through of runs to failure is higher in weaker banks across all sample periods.

Interestingly, runs without failure are significantly more common in the pre-Federal Reserve sample. After the introduction of the Fed, most runs are associated with failure. This holds both in the 1920s sample and the Great Depression. Two changes could account for this shift. First, the introduction of the Fed reduced the incidence of panic runs on relatively healthy banks that generally did not translate into failure, but it could not prevent runs from closing insolvent banks. Second, the Fed's founding may have displaced the private-sector arrangements that allowed banks to survive runs.[32]

Taking stock, the evidence indicates that, while runs can occur for strong and weak banks, they are substantially more likely to occur in weak banks, consistent with theories of fundamental-based panic runs. Moreover, conditional on a run, a weak bank is much less likely to survive the run. Thus, the combination of these two findings implies that runs almost never trigger the failure of strong banks.

6.2 Bank Responses to Runs

Banks with strong fundamentals typically do not fail when subject to a run. What measures do banks take in response to runs? To answer this question, we exploit additional information in newspapers. We identify seven common responses described in newspapers:

(i) the bank accommodated withdrawals by paying depositors;

(ii) the bank borrowed from another bank;

(iii) the bank sent a public signal to reassure depositors, such as conspicuously delivering gold or cash to the bank;

(iv) bank owners or investors injected equity or provided a loan;

(v) partial suspension of convertibility;

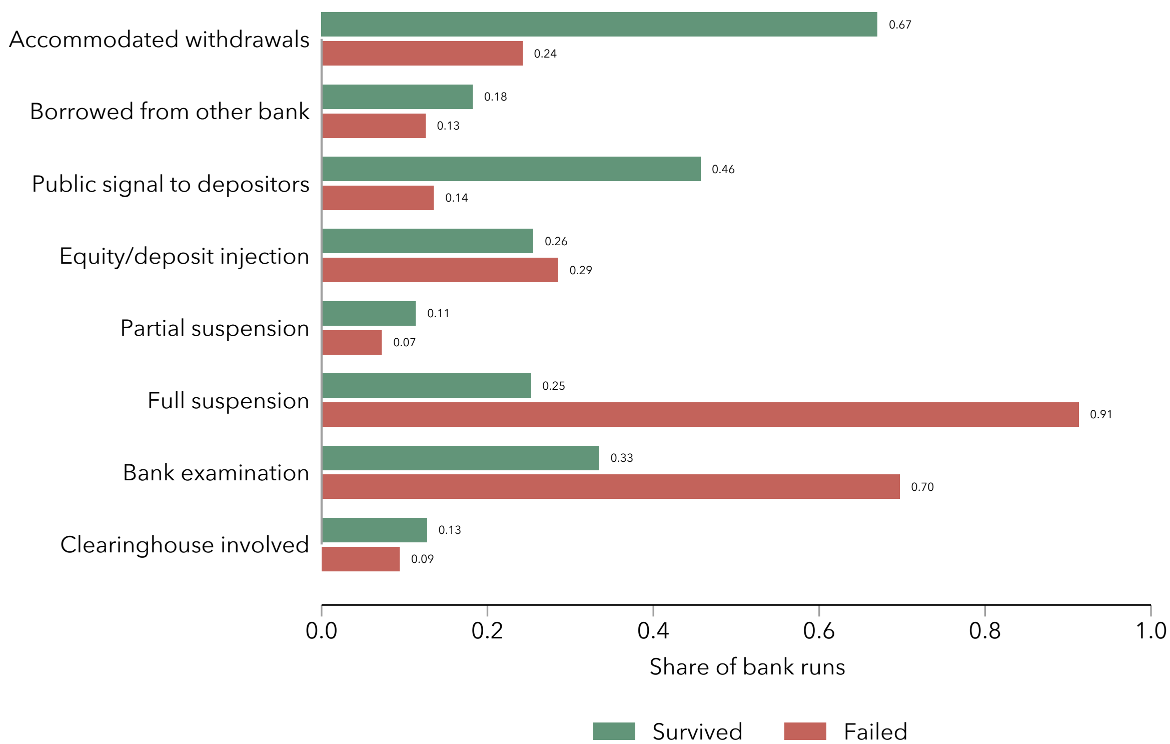
(vi) full suspension of convertibility; and

(vii) the bank underwent an examination by state supervisors, federal examiners, or by the local clearinghouse to ascertain its solvency.

[32] Bank failures are especially highly predictable in the post-1913 period. Moreover, bank asset quality and recovery rates are very low in this sample (Correia, Luck and Verner, 2025). This suggests that the Fed may have prevented runs on solvent banks and even delayed runs on insolvent banks via discount window access (Schwartz, 1992), implying that once runs occurred, they usually happened in deeply insolvent banks, making failure inevitable.

These responses are not mutually exclusive. We query an LLM to identify whether the newspaper description of a bank run episode mentions any of the seven potential bank responses. Appendix C provides the exact prompts we use.

Figure 7: ***Bank Responses to Runs Discussed in Newspapers***



Notes: This figure plots the share of episodes involving a bank run in which the newspapers mention one of the listed actions. The shares are reported separately for runs without and with failure.

Figure 7 reports the share of run episodes mentioning each of these responses. We distinguish between runs with and without failure. Banks that survive runs are more likely to be described as accommodating withdrawals compared to banks that fail in the run. They are also more likely to send a public signal to reassure depositors. Just like the commonly invoked George Bailey, who as noted above fended off the run on his bank by a mix of persuasion and using his own private wealth, owners of solvent banks commonly undertook successful measures to calm down worried depositors, including persuasion, other public signals such as delivering a "truckload of cash", or injecting additional equity into the bank (although the latter is equally likely in both failing and surviving banks). Surviving banks are also more likely to borrow from other banks.[33]

[33]We note that borrowing from other banks is likely to be underreported in newspapers, as banks may not want to make this public to avoid the stigma of needing assistance from other banks.

Surviving banks often take several of these actions in conjunction to ward off a run (see Figure A.7). Surviving banks are also more likely to partially suspend convertibility, though this is much less common than full suspension of convertibility.

In contrast, newspapers are substantially less likely to report that banks that fail in a run take these actions. Instead, newspapers emphasize that these banks are likely to fully suspend convertibility of deposits into currency. Moreover, banks that fail are also more likely to be subject to an examination by state supervisors, federal examiners, or the local clearinghouse to evaluate if the bank is solvent. When these banks are deemed to be insolvent, they are closed. Finally, in about 10% of runs, newspapers explicitly mention clearinghouse involvement, either through the provision of liquidity (loan certificates), examination, or another measure.[34]

6.3 Non-Fundamental Runs and Bank Failures

Can non-fundamental runs trigger bank failures? To answer this question, we use additional information from the context provided in newspapers around bank runs to classify a subset of the runs in our dataset as "non-fundamental" runs. We define a run as "non-fundamental" when newspapers affirmatively describe that the run was triggered by a specific, discrete episode of misinformation or confusion that triggered withdrawals and the source explicitly indicates that the information reported was false. These non-fundamental runs are *not* triggered by adverse news about the bank, failures or distress of other banks, systemic panic, or local economic shocks. See the prompt in Appendix C.1 for the exact definition used in the classification.

We identify 53 runs on national banks that are classified as non-fundamental, indicating that such runs are uncommon. Figure C.7 provides examples of non-fundamental runs. For instance, the First National Bank of Minneapolis was subject to a run after newspapers mistakenly reported it had suspended, confusing it with the First National Bank of Memphis. In Los Angeles in 1910, a large crowd gathered to see a famous boxer, who was depositing his money at the Merchants National Bank. This led to a run on the bank when others misinterpreted the crowd as a run.[35]

[34]Table A.8 examines whether the pass-through of runs to failure is lower for banks that are likely to be part of a local clearinghouse. We identify cities with clearinghouses using data from Jaremski (2018). National banks in cities with a clearinghouse were more likely to survive a run, suggesting that clearinghouses mitigated the effect of runs. This effect may be explained by the ability of clearinghouses to provide emergency liquidity as well as to credibly examine and transmit information about member banks' health. Appendix E discusses several cases where clearinghouse intervention prevented runs from translating into failures through examination and liquidity provision, such as Fourth National Bank of New York in 1873 and Metropolitan National Bank in 1884.

[35]This episode could be classified as both a non-fundamental run and an information-based run.

Table 6: ***Pass-Through of Runs to Failure: Narrative Classification of Non-fundamental Runs***

Dependent variable	Failure in t		
	(1)	(2)	(3)
Run	0.38***		
	(0.051)		
Non-fundamental run		0.11***	0.14***
		(0.036)	(0.045)
Other run		0.40***	0.44***
		(0.053)	(0.054)
Strong fundamentals			-0.0094**
			(0.0038)
Strong fundamentals × Non-fundamental run			-0.14***
			(0.045)
Strong fundamentals × Other run			-0.32***
			(0.051)
Observations	272535	272535	272535
Mean dep. var.	0.0084	0.0084	0.0084

Notes: This table presents estimates of Equation (6). We define a bank to have "strong" fundamentals when $\text{Fundamentals}_{bt-1}$ is in the upper tercile of the historical distribution. Driscoll and Kraay (1998) standard errors are reported in parentheses with a bandwidth of three years to allow for residual correlation within and across banks in proximate years. *,**, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 6 studies the pass-through of non-fundamental runs to bank failure based on estimating

$$\text{Failure}_{bt} = \beta_0 + \beta_1 \text{Non-Fundamental Run}_{bt} + \beta_2 \text{Other Run}_{bt} + \epsilon_{bt}, \tag{6}$$

While the typical run in our sample is associated with a 38 pp higher probability of failure (column 1), this drops to 11 pp for non-fundamental runs. Thus, non-fundamental runs are much less likely to result in bank failure. Moreover, column 3 shows that strong banks (fundamentals in the upper tercile) subject to non-fundamental runs have a zero probability of failure (0%=14%-14%). Thus, non-fundamental runs, while rare, can sometimes trigger failure if they happen to affect weak banks and uncover deeper asset quality issues.[36]

[36]For instance, several newspapers reported that a "heavy run" on the First National Bank (Clovis, NM) in January 1924 was caused by a "false rumor." However, the run led to suspension and receivership of the bank after it revealed that the bank had doubtful assets. The OCC assessed that all its assets were of "doubtful" quality at suspension, and the depositor recovery rate was only 29%, indicating deep insolvency. In this sense, even a random run can be the trigger for closing an insolvent bank.

6.4 Adverse Public Signals, Runs, and Failures

In Section 5, we found that runs, especially those that do not result in failure, often happen in the context of negative news about the macro economy and the banking sector. Motivated by this insight, we next study whether runs result in failure when they happen in the context of adverse public signals. Table 7 presents results from estimating variants of the following specification

$$\begin{aligned} \text{Failure}_{bt} = \beta_0 + \beta_1 \text{Run}_{bt} + \beta_2 \text{Adverse signal}_{rt} \\ + \beta_3 \text{Run}_{bt} \times \text{Adverse signal}_{rt} + \epsilon_{bt}, \end{aligned} \tag{7}$$

where Adverse signal_{rt} is an aggregate or regional adverse public signal, such as a run on another bank in the same city. This specification asks whether the pass-through from runs to failure differs for runs occurring after adverse public signals that are not bank-specific.

Table 7: ***The Pass-Through of Runs to Failure: The Role of Adverse Public Signals***

Dependent variable	Failure in t				
	(1)	(2)	(3)	(4)	(5)
Run	0.38***	0.44***	0.27***	0.20***	0.25***
	(0.051)	(0.052)	(0.057)	(0.065)	(0.056)
Run on other bank in same city		0.010***			0.016***
		(0.0038)			(0.0044)
Run on other bank in same city × Run		-0.14***			-0.15***
		(0.035)			(0.031)
Local business failure rate			0.78***		0.28**
			(0.16)		(0.13)
Local business failure rate × Run			7.03***		3.76***
			(1.36)		(1.10)
Business failure rate				1.28***	1.00**
				(0.34)	(0.43)
Business failure rate × Run				10.6***	7.11***
				(2.53)	(2.26)
Observations	282082	282082	243908	246960	243908
Mean dep. var.	0.0082	0.0082	0.0091	0.0090	0.0091

Notes: This table presents results from estimating variants of Equation (7). Driscoll and Kraay (1998) standard errors are reported in parentheses with a bandwidth of three years to allow for residual correlation within and across banks in proximate years. *,**, and *** indicate significance at the 10%, 5%, and 1% level, respectively.

Table 3 showed that the occurrence of a run in the same city is one of the strongest non-bank-specific antecedents of a run. Column 2 in Table 7 shows that runs that coincide with a run on another bank in the same city are 14 pp *less* likely to result in failure compared to other runs. The lower probability of failure for such runs is consistent with the notion that a run on another bank leads depositors to become "jittery." This finding is in line with depositors revising upward their belief that their own bank is insolvent. In some cases, this signal leads depositors to mistakenly run on healthy banks. As a consequence, runs within the context of runs on other banks are less likely to lead to failure. However, the response is not necessarily irrational, as these runs still involve a probability of failure of 30%, suggesting many of these banks that are run are indeed troubled.

In column 3 of Table 7, we use the state-level nonfinancial business failure rates as the proxy of an adverse signal from a deterioration in economic conditions. In contrast to the finding for runs on other banks, we find that runs that occur when the state-level business failure rate is elevated are *more* likely to be associated with a bank failure. In terms of magnitudes, a 1 pp rise in the state-level business failure rate—roughly a one standard-deviation increase—raises the pass-through from runs to failure by 7 pp. A rise in state-level business failures signals an increase in bank asset losses and thus an increased risk of bank insolvency. Runs in response to this deterioration of economic fundamentals are more likely to translate to failure, as these are precisely the times when banks are more likely to be insolvent. Columns 4 and 5 reveal a similar pattern when we use the aggregate nonfinancial business failure rate, rather than the local failure rate.

To further illustrate that bank runs and failures often follow a rise in non-financial business failures, Figure A.8 shows the dynamics of the state-level non-financial business failure rate around innovations in the state-level rate of bank runs and bank failures. State-level non-financial business failures are from *Bradstreet's* and are available quarterly. Both bank runs and bank failures are preceded by and coincide with an elevated rate of business failures, which lead to asset losses for banks. The relation is especially strong for bank failures. This evidence supports the key role of deteriorating fundamentals for understanding bank failures and runs. The small spike in business failures around runs without failure further suggests that increases in business failures can lead to runs on healthy banks that do not ultimately fail, consistent with information-based theories of runs.

7 Consequences of Runs for Lending and Real Activity

In this final section, we study the consequences of runs and failures on financial intermediation and real economic activity. We first study the impact of bank runs on surviving banks at the bank level. We then turn to the impact of bank runs and failures on local aggregate lending, deposits, and manufacturing activity.

7.1 Bank-Level Consequences of Bank Runs Without Failure

What are the consequences of bank runs without failure at the bank level? To answer this question, we estimate local projections of the following form:

$$\frac{Y_{bt+h} - Y_{bt-1}}{\text{Assets}_{bt-1}} = \alpha_b + \gamma_t + \beta^h \text{Run}_{bt}^{Without\ failure} + \epsilon_{bt+h}, \quad h = -5, \ldots, 4 \tag{8}$$

with the prediction horizon h running from five years before to five years after the distress episode. $\text{Run}_{bt}^{Without\ failure}$ is an indicator variable equal to one for a bank run without failure. We focus on runs without failure to understand the consequences of runs for surviving banks; for failing banks, we cannot observe bank-level outcomes after the run. We measure the change in a given bank balance sheet item ($Y_{b,t+h} - Y_{b,t-1}$) relative to total assets in the year before the event, $\text{Assets}_{b,t-1}$, so magnitudes are relative to pre-event total assets. α_b is a bank fixed effect, and γ_t is a year fixed effect.

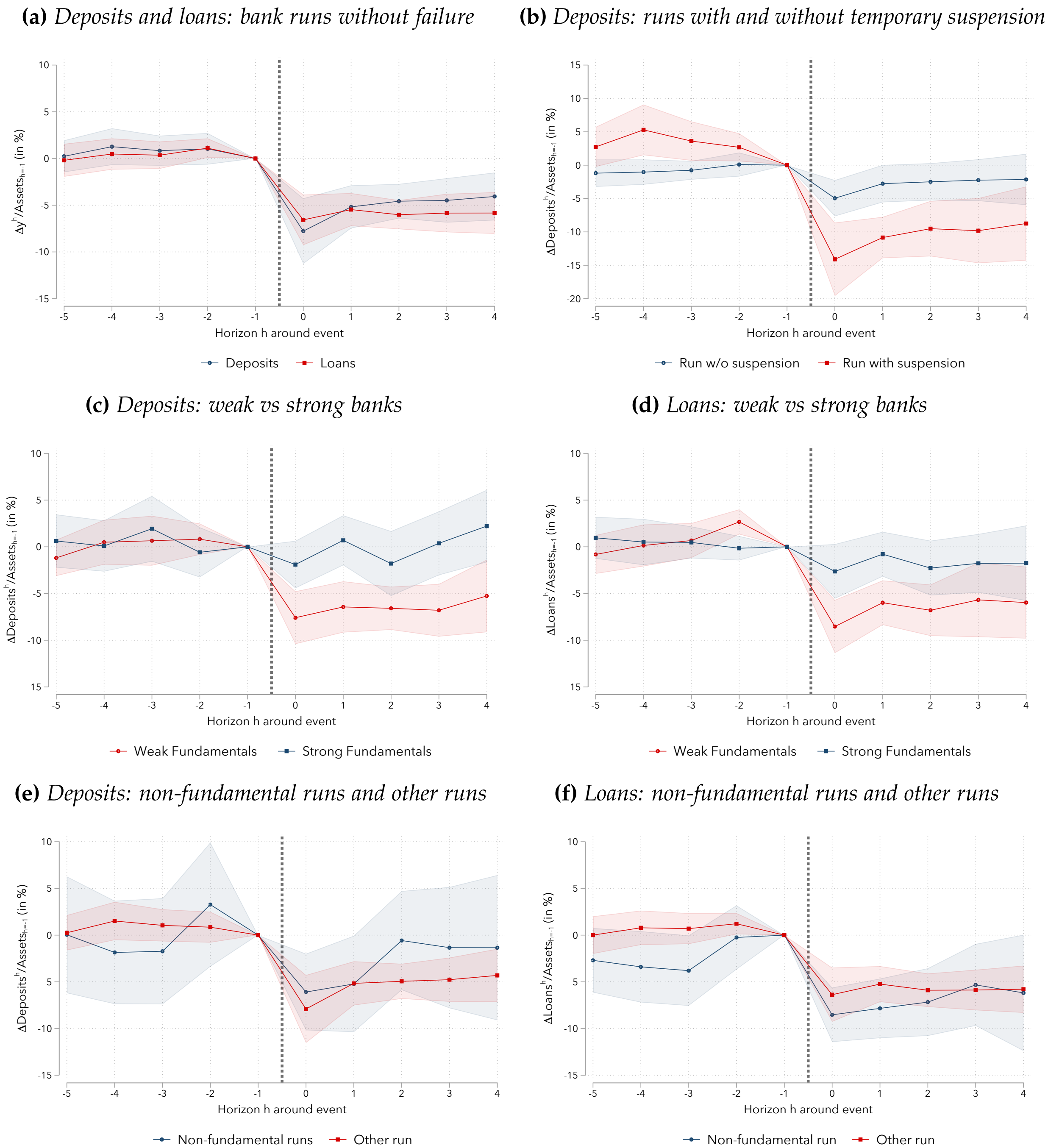
Panel (a) of Figure 8 shows the results from estimating Equation (8) for deposits and loans on the 1865-1934 sample of national banks. There is no pre-trend before a run without failure in either deposits or loans.[37] Once a bank is exposed to a run, deposits decline by around 7.5% as a share of pre-event assets and remain depressed thereafter. Loans decline by a similar magnitude. While we have selected episodes of bank distress that do not result in failure, these are still associated with a decline in deposits and loans. Thus, runs have a lasting impact at the bank level even in the absence of bank failure.

Panel (b) of Figure 8 further explores heterogeneity across runs with and without (temporary) suspension. Run episodes in which the bank is subject to a run but does not suspend at any point have relatively mild reductions in deposits of about 5%. After these episodes, banks recover their deposits within five years. Episodes with a run, suspension, and reopening are more severe, leading to a 15 percent decline in deposits that persists for five years. The larger decline for runs with suspension is likely because these runs are more severe. In addition, suspension itself may reduce depositor confidence and lead depositors to switch banks.

Dynamics For Weak and Strong Banks As discussed in Section 2, if runs themselves are the primary cause of distress, then runs should typically be associated with severe outcomes, irrespective of initial bank fundamentals. In contrast, if runs themselves are mainly a trigger of distress for weak banks, then runs should be less costly for banks with sound fundamentals. Thus, we next ask: What are the implications of a run

[37] As discussed above, for failing banks there is a gradual deterioration in loans and deposits before the run and failure (see Figure A.5 and the discussion in Appendix A.1).

Figure 8: ***Bank-Level Deposit and Loan Dynamics around Bank Runs Without Failure***



(a) *Deposits and loans: bank runs without failure*

(b) *Deposits: runs with and without temporary suspension*

(c) *Deposits: weak vs strong banks*

(d) *Loans: weak vs strong banks*

(e) *Deposits: non-fundamental runs and other runs*

(f) *Loans: non-fundamental runs and other runs*

Notes: This figure reports coefficients from estimating Equations (8) (panels a and b), (9) (panels c and d), and (10) (panels e and f). All panels focus on episodes where there is a run without bank failure. Shaded areas represent 95% confidence intervals based on Driscoll and Kraay (1998) standard errors with a bandwidth of three years to allow for residual correlation within and across banks in proximate years. Strong (weak) fundamentals are defined as $\text{Fundamental}_{b,t-1}$ being in the upper (lower) tercile.

without failure across weak and strong banks? To answer this question, we estimate local projections of the form:

$$\frac{Y_{b,t+h} - Y_{b,t-1}}{\text{Assets}_{b,t-1}} = \alpha_b + \gamma_t + \beta^h_S \text{Run}^{Without\ failure}_{b,t} \times \text{Strong Fundamentals}_{b,t-1} \tag{9}$$
$$+\beta^h_W \text{Run}^{Without\ failure}_{b,t} \times \text{Weak Fundamentals}_{b,t-1}$$
$$+\epsilon_{b,t+h}, \qquad h = -5, \ldots, 4$$

where strong (weak) fundamentals are defined as being in the top (bottom) tercile of the fundamentals measure in $t-1$.

Panels (c) and (d) of Figure 8 present the results. Runs without failure lead to larger deposit declines for banks with weak fundamentals. Banks with strong fundamentals see essentially no decline in loans and deposits, whereas banks with weak fundamentals see a persistent decline in deposits and lending of about 7%. Thus, bank fundamentals not only determine whether runs trigger bank failure; they also shape the consequences of bank runs when banks survive runs.

Taken together, the dynamics of deposits and loans around bank runs reinforce the notion that runs tend to be associated with severe distress when they happen in banks with weak fundamentals. Runs on banks with sound fundamentals have little impact on a bank's long-term viability.

Dynamics Around Non-Fundamental Runs A concern with studying the dynamics of deposits and loans around runs without failure is that such runs do not happen randomly but occur in response to bad news about the economy that leads depositors to run as a precautionary measure. These shocks can simultaneously reduce the demand for loans and deposits. To make some progress on the causal effect of runs, we replace the shock with our measure of non-fundamental runs discussed in Section 6.3. These runs, caused by false rumors or misinformation, are more exogenous events not caused by bank or economic conditions. This allows us to trace the effects of runs that are closer in spirit to the notion of random runs in Diamond and Dybvig (1983).

Figure 8 panels (e) and (f) study the consequences of non-fundamental runs and other runs by estimating

$$\frac{Y_{bt+h} - Y_{bt-1}}{\text{Assets}_{bt-1}} = \alpha_b + \gamma_t + \beta^h_{NF} \text{Run}^{Without\ failure,\ non-fund}_{bt} \tag{10}$$
$$+ \beta^h_{Other} \text{Run}^{Without\ failure,\ other}_{bt} + \epsilon_{bt+h}, \quad h = -5, \ldots, 4.$$

Non-fundamental runs without failure lead to a 5% drop in deposits and loans, which is slightly milder than for other runs without failure. Interestingly, deposits of banks subject to non-fundamental runs recover fully within two years of the run, similar to "run w/o suspension" events (see panel b). Loans remain roughly 5% lower three years after the run. Thus, runs that we specifically identify as non-fundamental and that are thus exogenous to bank and local economic conditions do lead to moderate disruptions in bank-level deposits and credit.

7.2 Local Consequences of Runs and Failures

We next examine the consequences of bank runs at the local aggregate level. As we have seen, most banking crises from 1863 to 1934 had a strong regional component. We can use our detailed data to exploit cross-regional variation to understand the local consequences of banking sector distress. We first examine the impact on city-level aggregate bank deposits and lending. We then provide evidence for the effect of runs on city-level manufacturing activity.

7.2.1 Deposits and Lending

Runs on weak vs strong banks We study the local economic consequences of bank runs on local deposits and lending, distinguishing between whether runs affect weak or strong banks. We estimate impulse responses from local projections of the following form:

$$g^{y}_{c,t-1,t+h} = \alpha_c + \beta^h_S \text{Run}^{Strong}_{c,t} + \beta^h_I \text{Run}^{Interm}_{c,t} + \beta^h_W \text{Run}^{Weak}_{c,t} \\ + X_{c,t-1}\Gamma^h + \epsilon_{c,t+h}, \quad h = 0, ..., H. \tag{11}$$

The outcome variable, $g^{y}_{c,t-1,t+h}$, is the symmetric growth rate of a variable such as total lending in city c from year $t-1$ to $t+h$, defined as $g^{y}_{c,t-1,t+h} = 100\frac{y_{c,t+h} - y_{c,t-1}}{0.5(y_{c,t+h} + y_{c,t-1})}$. The symmetric growth rate is bounded between -200 and 200. We employ this definition of growth to accommodate the possibility that an outcome in a local area goes to zero.[38] Outcome variables are constructed based on aggregating national bank balance sheets to the city and year level. Note that if a bank fails and exits, its loans and deposits go to zero. In estimating (11), we therefore weight by the number of banks in a city to mitigate the influence of large relative changes in small cities with few banks.[39]

[38] Results are similar if we use the log of $y_{c,t}$ as the outcome, but we prefer the symmetric growth rate to avoid exit and reentry of mostly smaller cities in the estimation sample.

[39] Figure A.9 reports the unweighted version of this specification. The impulse responses are qualitatively similar but quantitatively larger for the unweighted regression.

The shocks—$\text{Run}_{c,t}^{Strong}$, $\text{Run}_{c,t}^{Interm}$, and $\text{Run}_{c,t}^{Weak}$—capture local exposure to runs on banks with strong, intermediate, and weak fundamentals. These measures are defined as indicator variables that equal one if a bank with fundamentals in category j is subject to a run in city c and year t, and zero otherwise. Fundamentals are defined as the top (strong), middle (intermediate), and bottom (weak) terciles of the fundamentals measure, $Fundamentals_{b,t-1}$. By using indicator variables, we effectively exploit the extensive margin of the variation in whether a city is exposed to a run; results are qualitatively similar when using asset-weighted exposure to capture the share of banks subject to runs. In $X_{c,t-1}$, we include $L = 3$ yearly lags of both the run shocks and the dependent variable, as is standard in the local projections framework (Jordà, 2005). The local projection impulse responses are the sequence of estimated coefficients $\{\hat{\beta}_S^h, \hat{\beta}_I^h, \hat{\beta}_W^h\}$.

Figure 9 presents the impulse responses of city-level deposits (panel a) and loans (panel b) to innovations in the city-level runs on weak and strong banks. Runs predict a reduction in city-level deposits and loans, but only when they occur in weak banks. Runs on weak banks lead to a reduction in city-level deposits and lending of over 20% in the years immediately after the shock. In contrast, runs on strong banks are not associated with significant declines in deposits and lending. The difference between the responses for runs on strong and weak banks is statistically significant. This is in line with the view that runs themselves are not universally associated with severe crises but rather result in severe economic consequences when bank fundamentals are weak.

Non-fundamental and other runs In Figure 8, we saw that non-fundamental runs led to modest, but not negligible, reductions in bank-level deposits and lending. Figure 9 panels (c) and (d) report the implications of such runs for city-level deposits and lending based on estimating:

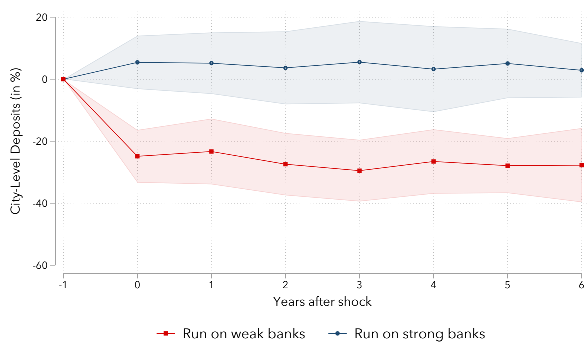
$$\begin{aligned} g^y_{c,t-1,t+h} = \alpha_c + \beta^h_{NF}\text{Run}_{c,t}^{Non-fund} + \beta^h_{Other}\text{Run}_{c,t}^{Other} \\ + X_{c,t-1}\Gamma^h + \epsilon_{c,t+h}. \quad h = 0, ..., H. \end{aligned} \tag{12}$$

$\text{Run}_{c,t}^{Non-fund}$ is an indicator for whether there is a non-fundamental run in city c and year t as defined in Section 6.3 and $\text{Run}_{c,t}^{Other}$ is an indicator for the occurrence of other runs in a city. While non-fundamental runs reduce deposits and lending at affected banks, Figure 9 shows they have essentially no effect on local deposits and lending.[40] The

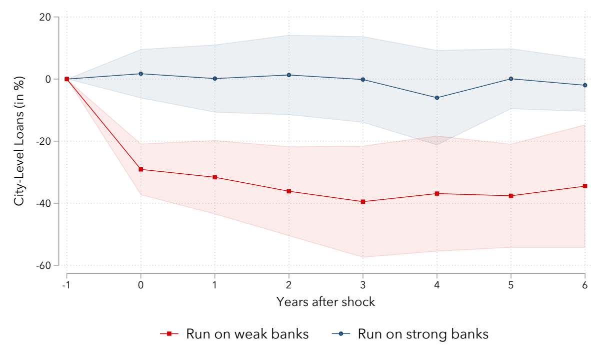
[40]Loans do exhibit a gradual decline, but the estimates are noisy and not statistically significant. In the unweighted specification (Figure A.9, panel d), the impact of non-fundamental runs on loans is more precise and close to zero.

Figure 9: ***Consequences of Runs for City-Level Deposits and Lending***

(a) *Impact of runs on deposits, weak vs strong banks*

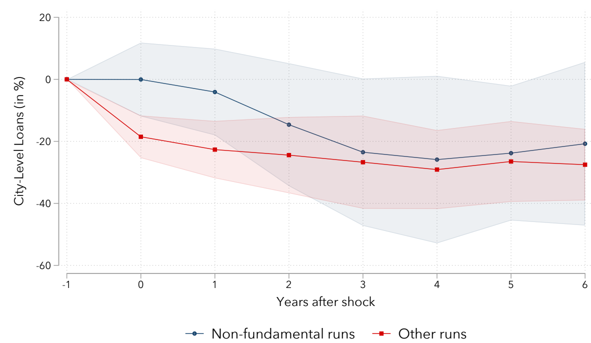


(b) *Impact of runs on loans, weak vs strong banks*



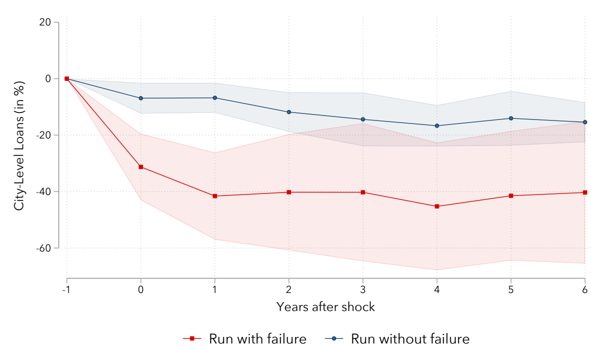
(c) *Impact of runs on deposits, non-fundamental runs*

(d) *Impact of runs on loans, non-fundamental runs*



(e) *Impact of runs with and without failure on deposits*

(f) *Impact of runs with and without failure on loans*



Notes: This figure presents impulse responses based on estimation of Equation (11) (panels a and b), Equation (12) (panels c and d), and Equation (13) (panels e and f). The analysis is based on a city-level annual panel. Loans, deposits, and bank distress are based on aggregating the sample of national banks from 1863-1934 to the city level at an annual frequency. The regressions are weighted by the number of banks in a city; unweighted estimates are reported in Figure A.9. Error bands represent 95% confidence intervals based on Driscoll and Kraay (1998) standard errors with a horizon-dependent bandwidth of $L_h = \lceil 1.5h \rceil$.

null aggregate effect is because deposits are reallocated seamlessly to other local banks following non-fundamental runs.

Runs with and without failure Next, we study the consequences of runs with and without failure. This provides another test to address whether runs alone can be associated with severe economic consequences, or whether bank failures, indicative of bank insolvency, are necessary for runs to be associated with severe crises.

To do this, we estimate impulse responses from the following local projections specification:

$$g^{y}_{c,t-1,t+h} = \alpha_c + \beta^h_{RNF}\text{Run}^{Without\ failure}_{c,t} + \beta^h_{RWF}\text{Run}^{With\ failure}_{c,t} \\ + X_{c,t-1}\Gamma^h + \epsilon_{c,t+h}, \quad h = 0, ..., H. \tag{13}$$

$g^{y}_{c,t-1,t+h}$ is again the symmetric growth rate of aggregate city-level loans or deposits from $t-1$ to $t+h$. We consider two mutually exclusive bank distress episodes: (i) run without failure, and (ii) run with failure. Each measure is defined as an indicator variable for whether a city is subject to a given distress episode in year t. This exercise allows us to distinguish the consequences of episodes that are closer to pure liquidity events from those involving bank insolvency. Given that the likelihood of failure conditional on a run is strongly predictable based on bank fundamentals, this exercise complements the previous analysis on runs across banks with strong and weak fundamentals. However, instead of using an *ex ante* measure of bank health, here we use an *ex post* measure of bank distress.

Panels (e) and (f) in Figure 9 present the results from estimating Equation (13). Not all bank distress episodes are alike. Runs with failure translate into substantially larger contractions in city-level deposits and lending than runs without failure. Runs without failure lead to a modest decline in city-level deposits that is borderline statistically significant. The effect on loans is more persistent and significant, but also modest in size. Thus, bank distress involving bank failure is associated with substantially worse lending contractions than episodes with runs that do not result in failure. Failures, which are related to bank insolvency, are necessary for bank runs to involve severe local deposit and lending contractions.

In terms of magnitudes, Figure 9 shows that if a city experiences a run that results in bank failure, deposits decline by around 30% and loans by 40%. The responses are persistent. These responses are large in part because of restrictions on branching and acquisitions, which meant that bank failure would lead to persistent reductions in banking services in a city (Quincy and Xu, 2025). This is especially true in smaller cities with fewer banks. In contrast, if a city is exposed to a run that does not result in bank failure, both deposits and loans decline by about 10%.

7.2.2 Manufacturing Activity

As a final exercise, we study the real economic consequences of runs at the local city level. We use a newly collected dataset on weekly local economic activity constructed from *Bradstreet's*, a business journal that provides city-level summaries of economic conditions in major cities for various sectors, including manufacturing, retail trade, and wholesale trade.[41] The key benefit of this data is that it is weekly, so we can match to runs at the weekly level and study high-frequency real dynamics. We focus on *Bradstreet's* description of activity in the manufacturing sector. Using the short textual descriptions, we classify economic conditions into poor (1), fair (2), or good (3). We normalize our index of manufacturing activity to have the same variance as the cyclical component of industrial production (IP), defined as the deviation of log IP from its one-sided three-year moving average. This analysis covers 30 major cities from 1917 through 1935, providing high-frequency evidence on the consequences of runs for the real economy.

Before proceeding, we note that the results based on our index of manufacturing activity from *Bradstreet's* are mainly driven by larger cities during the Great Depression. Moreover, the index is based on qualitative information, so the transformation to a numeric index may introduce measurement error. Nevertheless, in the appendix, we show that the aggregated city-level manufacturing index constructed from *Bradstreet's* is highly correlated with the cyclical component of industrial production. It captures the major 1920-21 recession, the mild recessions of 1923-24 and 1926-27, and the major downturn starting in August 1929, indicating that it contains valuable information at the business cycle frequency.

Runs on weak vs strong banks We first study the city-level consequences of runs on weak versus strong banks. We estimate impulse responses to run events using the specification:

$$\begin{aligned} Y^{Manuf}_{c,t+h} &= \alpha_c + \beta^h_S \text{Run}^{Strong}_{c,t} + \beta^h_I \text{Run}^{Interm}_{c,t} + \beta^h_W \text{Run}^{Weak}_{c,t} \\ &\quad + X_{c,t-1}\Gamma^h + \epsilon_{c,t+h}, \quad h = 0, ..., H. \end{aligned} \tag{14}$$

$Y^{Manuf}_{c,t}$ is the city-level manufacturing index in city c and week t. α_c is a city fixed effect. The shocks $\text{Run}^{Strong}_{c,t}$, $\text{Run}^{Interm}_{c,t}$, and $\text{Run}^{Weak}_{c,t}$ are indicator variables for whether a strong, intermediate, or weak bank was subject to a run in city c and week t based on the tercile of

[41]The underlying data are weekly for 1917 through March 11, 1933, and monthly for the remainder of 1933 through 1935. For the latter period, we fill values forward to retain a weekly panel structure. See Appendix D.1 for details on the *Bradstreet's* data and the construction of the manufacturing activity index.

$Fundamentals_{b,t-1}$. We include 24 weekly lags of both the dependent and the innovation variables in $X_{c,t-1}$ and study responses out to 80 weeks (1.5 years).

Figure 10a shows that runs lead to substantially worse declines in city-level manufacturing when they occur in weak banks. Runs on strong banks do not predict a deterioration in city-level manufacturing. The estimates are slightly positive, close to zero, and not statistically significant at most horizons. In contrast, runs on weak banks lead to a gradual decline in manufacturing activity over the subsequent year. Recall that the manufacturing index is normalized to have the same mean and variance as the cyclical component of aggregate industrial production. Thus, in terms of magnitudes, runs on weak banks in a city are associated with a roughly 5% decline in city industrial production after one year.

We stress one important point about identification that affects the interpretation of the estimates of Equation (14) in Figure 10. The negative response of real activity following a run may be caused by the run itself, but it can also be caused by an underlying real economic shock that leads depositors to run. While the high-frequency nature of our analysis allows us to closely link the timing of runs and real activity, we cannot fully disentangle these stories. However, distinguishing between runs on weak versus strong banks (and runs with versus without failure below) allows us to understand which types of runs are associated with the most severe economic consequences. The findings imply that runs alone, without fundamental weaknesses or failure, do not lead to severe negative real effects.

Runs with and without failure Finally, we study the city-level consequences of runs with and without failure for manufacturing activity. We estimate impulse responses from the following local projection at the weekly level.

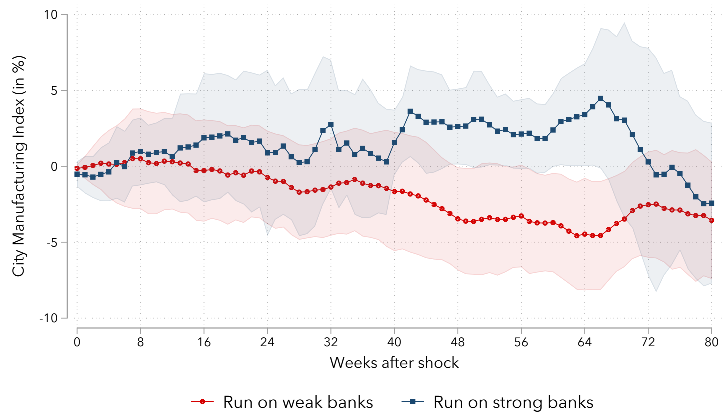
$$\begin{aligned} Y^{Manuf}_{c,t+h} = \alpha_c + \beta^h_{RNF}\text{Run}^{Without\ failure}_{c,t} + \beta^h_{RWF}\text{Run}^{With\ failure}_{c,t} \\ + X_{c,t-1}\Gamma^h + \epsilon_{c,t+h}, \quad h = 0, ..., H. \end{aligned} \tag{15}$$

In this specification, $\text{Run}^{Without\ failure}_{c,t}$ is an indicator for whether a city c is subject to a run without failure in week t, and $\text{Run}^{With\ failure}_{c,t}$ is an indicator for whether there is a run with a bank failure. We define a run as involving failure if the bank fails within 30 days of the date of the run, based on the evidence that most failures occur within 30 days of the run (see Figure 6).

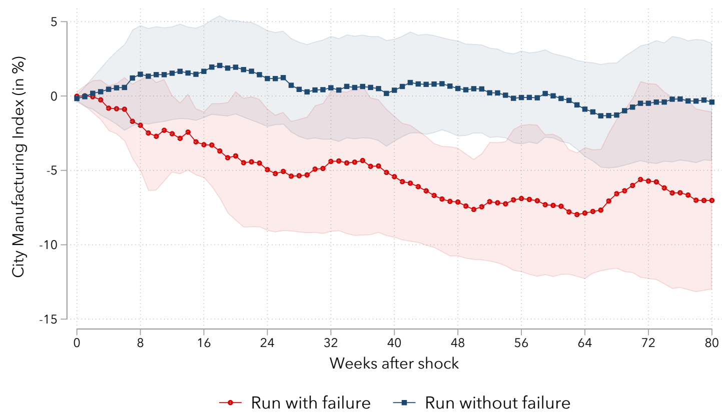
Figure 10b reveals that runs with bank failures lead to substantially worse declines in local economic activity than runs without failure. The response of manufacturing

Figure 10: ***Consequences of Runs for City-Level Manufacturing Activity***

(a) *Impact of runs on manufacturing index, weak vs strong banks*



(b) *Impact of runs with and without failure on manufacturing index*



Notes: This figure presents impulse responses based on estimation of Equation (14) (panel a) and Equation (15) (panel b). The analysis is based on a city-level weekly panel from 1917 to 1935. The outcome variable is a city-level weekly index of manufacturing activity, which we normalize to have the same mean and variance as industrial production; see Appendix D.1 for details. Error bands represent 95% confidence intervals based on standard errors dually clustered on city and time (week).

activity to runs without failure (runs with failure) is similar to the response to runs on strong (weak) banks in panel (a). Runs without failure do not result in significantly lower manufacturing activity within the next 18 months. Runs with failure predict a decline in manufacturing activity of 5–8%.

Overall, the evidence on local consequences of runs for credit markets and real activity tempers the view that runs that originate as pure liquidity events can generate severe disruptions in local credit markets and real activity. Runs on strong banks and non-fundamental runs carry limited adverse effects beyond the affected banks. Instead, the evidence indicates that poor fundamentals are necessary for runs to generate severe local financial disruptions. A key channel through which runs disrupt real activity is through bank failure, which, in turn, usually requires that a bank is fundamentally insolvent, or at least very weak in terms of fundamentals (Correia, Luck and Verner, 2025). Thus, runs should not be seen as universally carrying high economic costs; rather, runs involve large economic contractions when bank fundamentals are particularly weak.

8 Conclusion

In this paper, we apply textual analysis and large language models to historical newspapers to study bank runs recorded in digitally available newspapers in the United States from 1863 to 1934. Our novel database uncovers nearly four thousand bank runs. We combine these data with balance sheet information on banks, allowing for the first comprehensive micro-level evidence on the causes and consequences of bank runs. Our data allow us to derive several key insights about the nature of banking crises.

We provide three main findings. First, runs are more likely in banks with observably weaker fundamentals, broadly consistent with fundamental-based panic runs. But strong banks can also be run, especially in response to adverse news about the economy or the banking system, in line with information-based theories of runs. Second, most runs do not result in bank failure, and runs typically only trigger the failure of weak banks. Banks with strong fundamentals manage to survive runs through various mechanisms, including interbank cooperation, signaling strength, equity injections or loans, examination, and suspension, consistent with information-based theories of runs. Third, weak fundamentals are necessary for bank runs to result in substantial contractions in financial intermediation and real manufacturing activity. Moreover, bank failure is an important mechanism for capturing the local economic costs of runs.

Taken together, our findings lend little support to the view that small shocks can result in large jumps to bad equilibria via runs on demandable debt. In contrast, our findings suggest that absent government interventions in the banking sector such as deposit insurance, runs can occur in both weak and strong banks, but poor fundamentals are necessary for runs to result in bank failures. Moreover, poor bank fundamentals are necessary for runs to result in severe real economic contractions. Failures are a key proxy for economic distress resulting from turmoil in the banking sector. Events that are primarily liquidity driven are less commonly associated with severe economic outcomes and should be less of a concern than solvency-driven crises.

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